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Some diophantine equations with two S-units and one S-integer unknown

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1 Introduction

Dipohantine equations are equations of which we search integral solutions. The simplest case is when the equation is defined over $\mathbb{Q}$ and the unknowns must take integer values.

More generally, let k be a number field and S be a finite set of places of k containing the archimedean ones. We can consider equations defined over k and the unknowns to take values in the set of S -integers (O_S), see Definition 2.1. In the case $k = \mathbb{Q}$, the set S is formed by the usual absolute value and a finite number of primes. The S -integers are those rational numbers whose denominator only contains primes from S .

In 2.1 we also define the S -units (O_S^*). In the case $k = \mathbb{Q}$ they are those rational numbers whose numerator and denominator only contain primes from S . To get meaningful results we sometimes only consider solutions in S -units. For example we have the S -unit equation

$$x + y = 1 \quad (x, y) \in O_S^* \times O_S^* \quad (1)$$

It only has finitely many solutions [HS00] §D.8.

A fundamental result about diophantine equations is Siegel's theorem [HS00] §D.9 or [Zan03] III§2.

Theorem 1.1. (Siegel). *Let C/k be an affine irreducible algebraic curve embedded in k^n . Suppose that the set*

$$C(O_S) = \{(x_1, \dots, x_n) \in C \mid x_i \in O_S, i = 1, \dots, n\}$$

is infinite. Then C has genus zero and at most two points at infinity.

As an example we can deduce that the S -unit equation (1) only has finitely many solutions from Siegel's theorem. Consider the affine irreducible algebraic curve $\mathbb{P}^1(k) \setminus \{0, 1, \infty\}$. It can be embedded in k^3 as the intersection of $XY = 1$ and $(1 - X)Z = 1$. It has three points at infinity. Hence, by Siegel's theorem one only has finitely many solutions for

$$x_1 x_2 = 1 \quad \text{and} \quad (1 - x_1) x_3 = 1 \quad (x_1, x_2, x_3) \in O_S^3 \quad (2)$$

For every solution (x_1, x_2, x_3) of (2) we have a solution $(x_1, 1 - x_1)$ of (1). Vice versa, for every solution (x, y) of (1) we have a solution $(x, 1/x, 1/y)$ of (2). Then the S -unit equation only has finitely many solutions because (2) only has finitely many of them.

1.1 The equation $f(x, y) = z$

Siegel's theorem is a powerful tool, however it can only be applied to curves. The equations we study in this thesis are defined by surfaces instead, in fact, they contain three unknowns. More precisely, the equations we consider all have the shape

$$f(x, y) = z \quad (x, y, z) \in O_S^* \times O_S \times O_S^* \quad (3)$$

for some polynomial $f \in k[X, Y]$. If f is a homogeneous polynomial with respect to some weights, a classical result of Mahler (Theorem I.7 p. 12 of [Zan03]) can be used to understand when it has finitely many solutions. However, here we do not assume f homogeneous. With some generic hypothesis on f and some additional conditions on x and z we establish that these equations only have finitely many solutions.

We remark that if one of the variables x , y or z is fixed, then we are left with only two unknowns, and it is well known how to deal with equations of the resulting form. If y is fixed we have (a particular instance of) the equation

$$g(x, z) = 0 \quad x, z \in O_S^*$$

which has only finitely many solutions unless g is very special: $X^h - \eta Y^k$ or $\eta X^h - Y^k$ must divide g for some $h, k \in \mathbb{N}$ and $\eta \in \bar{\mathbb{Q}}^*$. This is proved in Lemmas 2.17 and 2.18. If instead z (or equivalently x) is fixed we have the more general equation

$$g(x, y) = 0 \quad x \in O_S^*, y \in O_S$$

and using Siegel's theorem one can see that it can have infinitely many solutions only if g is, again, very special. There must be a positive integer h and a polynomial $p \in \bar{k}[X, Y]$ such that $g(X^h, p(X, X^{-1})) = 0$ (Ex. 10 p. 43 of [Zan03]).

The case where $x = a^n$ and $z = b^m$ are powers of given integers, and a and b are not coprime has been studied by P. Corvaja and U. Zannier in [CZ00]. Our first result is a natural generalization to number fields of the Main Theorem of [CZ00].

Theorem 1.2. *Let $f(X, Y) = a_0(X)Y^d + a_1(X)Y^{d-1} + \dots + a_d(X)$ be a polynomial of degree $d \geq 2$ in Y with coefficients in a number field k . Let a and b be any two nonzero integers of k , not roots of unity. Suppose that*

- (i) a_0 is constant
- (ii) the equation $f(0, Y)$ has no repeated roots
- (iii) there exists an absolute value $|\cdot|$ of the number field k such that $|a| < 1$ and $|b| < 1$

If the equation

$$f(a^m, y) = b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$, such that $\min(m, n) \rightarrow \infty$, then there exist an integer $h > 0$ and a polynomial $p(X) \in \bar{k}[X]$ such that $f(X^h, p(X))$ has only one term. Also a, b are multiplicatively dependent.

We prove this in chapter 3 via the Schmidt subspace theorem, a deep result from diophantine approximation. The possibility of this extension to number fields was hinted in Remark 4 of [CZ00]. In Remark 1, they also noted that it could be extended to more general S -units. In chapter 4 we prove that this can be done in the following way. In Theorem 1.2 the powers a^m and b^n are substituted with S -units x and z respectively. Hypothesis (iii) must be changed into this one

(iii) $\max(|x|, |z|) \rightarrow 0$ for some absolute value $|\cdot|$ and the height of the S -units x and z are bounded, for some constant $c > 0$ by

$$h(x) < -c \log |x|$$

$$h(z) < -c \log |z|$$

The three technical hypotheses of Theorem 1.2 are important for our method. The next results aim at weakening hypotheses (ii) and (iii). We can drop (iii) if we restrict ourselves to a much smaller class of equations. For example in chapter 5 (see 5.6) we prove the following corollary where we only suppose a and b multiplicatively independent.

Corollary 1.3. *Let $u \in \mathbb{Q}[x]$ be a polynomial of degree greater than zero such that $u(0) = 0$, and let $d \geq 2$ be an integer. Fix a, b two multiplicatively independent integers. Then the equation*

$$y^d = u(a^m) + b^n$$

doesn't have an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times \mathbb{Z}$ with $\min(m, n) \rightarrow \infty$.

Hypothesis (ii) can also be weakened. We were able to find a result similar to the Main Theorem of [CZ00] supposing strictly less. The precise statement is contained in the Main Corollary 6.12 and requires looking at the Newton polygon of f . As simple corollaries we can prove 6.22 and 6.23. These are significant consequences since they generalize the Corollary of [CZ00] in two different directions.

1.2 Powers with few nonzero digits

The paper [CZ00] proved to have some important applications. An immediate consequence is, for example, the classification of perfect squares that expressed in a given base only have three nonzero digits (see Remark 2 of [CZ00]).

Theorem 1.4. *Let S be the set of all squares that in base 10 are written with exactly three non zero digits. All but finitely many squares in S are squares of a two digits number.*

Proof. Fix $a_0, a_1, a_2 \in [1, 9]$ and consider the squares of the form

$$y^2 = a_0 10^{n_0} + a_1 10^{n_1} + a_2 10^{n_2} \quad (4)$$

with $n_0 < n_1 < n_2$. 10^{n_0} is the greatest power of 10 which divides y^2 then $n_0 = 2t$ is even. If we put $y' = y/10^t$ then the previous equation becomes

$$y'^2 = a_0 + a_1 10^{n_1 - n_0} + a_2 10^{n_2 - n_0}$$

so that in equation (4) we can assume $n_0 = 0$. Consider the polynomial $f(X, Y) = (Y^2 - a_0 - a_1 X)/a_2$ and note that

$$f(10^{n_1}, y) = 10^{n_2}$$

If we have infinitely many squares of the form (4), Theorem 1.2 gives an identity $p(X)^2 = a_0 + a_1 X^h + a_2 X^k$ for a polynomial p and integers h, k . From this we find that $k = 2h$ or $h = 2k$ and $p(X) = c_0 + c_1 X^h$ or $p(X) = c_0 + c_1 X^k$ for some algebraic numbers c_0 and c_1 such that c_0^2, c_1^2 and $2c_0 c_1$ are integers and $0 < |c_0|, |c_1| \leq 3$. Moreover, from Remark 3.8 we get that for all but finitely many solutions of (4) there exists such a p and a rational number r such that $y = p(10^r)$. We can easily see that c_0, c_1 and r must be integers proving that y has only two nonzero digits. $\square$

Later, this was also proved in [Ben12] by means of a different technique. The problem of classifying powers with few non zero digits is of great interest and has been studied extensively. The previous result works for every base. For base 2 we can even go further. In [CZ13] the authors prove

Theorem 1.5. *There are only finitely many odd perfect powers in $\mathbb{N}$ having precisely four nonzero digits in their binary expansion.*

However, we do not know in general how to classify powers with four non zero digits.

1.3 Recent developments

Work on this subject has been carried on by A. Levin. In [Lev13] he considered the equation

$$f(u, v) = w \quad (u, v, w) \in O_S^{*3}$$

where the cardinality of S is at most 3. Under generic assumptions on f he gives an effective procedure to compute all the solutions.

A. Levin is also studying how to weaken hypothesis (ii). His results (not published yet) are different from the ones we present here since they rely on a more general version of the subspace theorem where points on algebraic varieties are approximated by closed subschemes. Instead, our point of view is more analytical.

It is not known how to deal without hypothesis (ii). Also we do not know if hypothesis (iii) can be removed. For example we get from Theorem 1.2 that there are only finitely many solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times \mathbb{Z}$ for the equation

$$y^2 = 2^n + 6^m + c$$

where $c > 0$ is any fixed parameter. However, the same is unknown for the equation

$$y^2 = 2^n + 3^m + c$$

which is conjectured to have only a finite number solutions (see Ex. 13 p. 61 of [Zan03] and a conjecture by Lang and Vojta [HS00] p. 486). In the special case $c = 0$ though, there is not an infinite sequence of solutions with $\min(m, n) \rightarrow \infty$ (use, for example, 1.3). The case $c = 1$ has been studied by D. Leitner in [Lei11], he proves that the equation $y^2 = 2^n + 3^m + 1$ has only finitely many solutions. His method cannot be adapted for other values of c .

2 Preliminaries

We collect here some well-known results and a lemma that will be used throughout this thesis. Since they will frequently appear in the following chapters we believe it is convenient to state them once and for all at the beginning.

2.1 Basic definitions and properties

Let k be a number field. We denote by M_k the set of places of k and with $M_{k,\infty}$ the (finite) set of archimedean places of k . For a place v and $x \in k$ we write $|x|_v$ for the norm of x with respect to the place v .

Definition 2.1. Let S be a finite set of places of k containing the archimedean ones. We define the S -integers

$$O_{k,S} = \{x \in k \mid |x|_v \leq 1 \ \forall v \in M_k \setminus S\}$$

It is a ring. The invertible elements of $O_{k,S}$ form the group $O_{k,S}^*$ of S -units

$$O_{k,S}^* = \{x \in k \mid |x|_v = 1 \ \forall v \in M_k \setminus S\}$$

So for example if we take $S = M_{k,\infty}$ then $O_{k,S} = O_k$ is the ring of integers of k . We will sometimes write O_S , O_S^* , M_∞ for $O_{k,S}$, $O_{k,S}^*$ and $M_{k,\infty}$ respectively when the number field is clear from the context. Next, we recall some useful results. We refer to [CF67], II.

Theorem 2.2. (Roots of unity). *Let k be a number field. A number $x \in k$ is a root of unity ($x^n = 1$ for some $n > 0$) if and only if $|x|_v = 1$ for every place v of k .*

Theorem 2.3. (Product Formula). *Let k be a number field. For any $x \in k^*$ we have that $|x|_v \neq 1$ only for a finite set of places v and the following formula holds*

$$\prod_{v \in M_k} |x|_v = 1$$

Theorem 2.4. (Dirichlet). *Let k be a number field and S be a finite set of places of k containing the archimedean ones. The group $O_{k,S}^*$ is a finitely generated abelian group of rank $\#S - 1$.*

We also recall the concept of height of a point in $\mathbb{P}^n(k)$, it is important to formulate the subspace theorem. The greater the height, the greater the "complexity" of the point.

Definition 2.5. For a point $P = (x_0 : \cdots : x_n) \in \mathbb{P}^n(\mathbf{k})$ or $P = (x_0, \dots, x_n) \in \mathbf{k}^{n+1}$ we define the *Weil height* of P as

$$H(P) = \prod_{v \in M_{\mathbf{k}}} \max(|x_0|_v, \dots, |x_n|_v)$$

The *logarithmic height* is

$$h(P) = \log H(P)$$

If $x \in \mathbf{k}$ we define its height as $H(x) = H(1 : x)$.

Remark 2.6. The height is well defined (independent from the choice of projective coordinates) thanks to the product formula.

2.2 Subspace Theorem

The subspace theorem is a result of diophantine approximation proved originally by W. M. Schmidt in 1972 ([Sch80], VI). It states that the points for which a product of linear forms is small enough with respect to the height are contained in a *finite* union of proper linear subspaces, hence the name subspace theorem. This is the main tool used in my thesis. There are various versions of the theorem, we recall two of them. The first one is Theorem 1.6 of [EE93], IV. We use the notation $|(x_1, \dots, x_n)|_v = \max(|x_1|_v, \dots, |x_n|_v)$.

Subspace Theorem 2.7. (Schlickewei). *Let $\mathbf{k}$ be a number field and S be a finite set of places of $\mathbf{k}$ containing the archimedean ones. For $v \in S$ let $L_{i,v}$, $i = 1, \dots, n$ be linearly independent linear forms in n variables, with algebraic coefficients, and let $\epsilon > 0$. Then the solutions $\mathbf{x} \in \mathbf{k}^n$ to the inequality*

$$\prod_{v \in S} \prod_{i=1}^n \frac{|L_{i,v}(\mathbf{x})|_v}{|\mathbf{x}|_v} \leq H(\mathbf{x})^{-n-\epsilon} \quad (5)$$

all lie in a certain finite union of proper linear subspaces of $\mathbf{k}^n$.

The previous version is somehow "projective" (in the sense that the quantities on both sides are invariant by linear equivalence). The next version can be very easily derived from the previous one. It is no longer "projective" but it has the advantage of having a more compact form. It is the version adopted in the discussion of the main results of this thesis.

Subspace Theorem 2.8. (Schlickewei). *Let k be a number field and S be a finite set of places of k containing the archimedean ones. For $v \in S$ let $L_{i,v}$, $i = 1, \dots, n$ be linearly independent linear forms in n variables, with algebraic coefficients, and let $\epsilon > 0$. Then the solutions $\mathbf{x} \in O_{k,S}^n$ with all the coordinates S -integers to the inequality*

$$\prod_{v \in S} \prod_{i=1}^n |L_{i,v}(\mathbf{x})|_v \leq H(\mathbf{x})^{-\epsilon} \quad (6)$$

all lie in a certain finite union of proper linear subspaces of k^n .

Theorem 2.8 can be derived from 2.7 simply by noting that if $\mathbf{x} \in O_{k,S}^n$ we have $H(\mathbf{x}) = \prod_{v \in M_k} |\mathbf{x}|_v \leq \prod_{v \in S} |\mathbf{x}|_v$ hence $\prod_{v \in S} \prod_{i=1}^n |\mathbf{x}|_v \geq H(\mathbf{x})^n$.

Remark 2.9. *The linear subspaces of k^n are the sets of points $\mathbf{x} = (x_1, \dots, x_n) \in k^n$ such that $A_1 x_1 + \dots + A_n x_n = 0$ where the A_i are in k .*

Remark 2.10. *When c is the coefficient of a linear form and $c \in \bar{k} \setminus k$ the absolute value $|c|_v$ appearing in (5) and (6) is not well-defined. We mean that we can choose any extension of v to $k(c)$.*

A corollary of Theorem 2.7 is the so-called generalized Roth's theorem, a generalization of the well-known Roth's theorem which is already quite a remarkable result.

Generalized Roth's theorem 2.11. *Let k be a number field and S be a finite set of places of k containing the archimedean ones. For $v \in S$, let α_v be algebraic over k and let $\epsilon > 0$. Then there exist at most finitely many numbers $\beta \in k$ such that*

$$\prod_{v \in S} \min(1, |\alpha_v - \beta|_v) \leq H(\beta)^{-2-\epsilon} \quad (7)$$

Proof. Fix any subset S' of S and consider the set of those β which satisfy (7) and such that for every $v \in S$

$$|\alpha_v - \beta|_v < 1 \Leftrightarrow v \in S' \quad (8)$$

We just need to prove that this set of β 's is finite.

We use Theorem 2.7 with $n = 2$ and S the one given in the statement of Theorem 2.11. Consider the linear forms

$$\begin{array}{lll} L_{1,v}(x_1, x_2) = x_1 & L_{2,v}(x_1, x_2) = \alpha_v x_1 - x_2 & \text{for every } v \in S' \\ L_{1,v}(x_1, x_2) = x_1 & L_{2,v}(x_1, x_2) = x_2 & \text{for every } v \in S \setminus S' \end{array}$$

We take the point $P = (1, \beta)$ and we estimate the product

$$\begin{aligned}
\prod_{v \in S} \frac{|L_{1,v}(P)|_v |L_{2,v}(P)|_v}{|P|_v^2} &= \prod_{v \in S} \frac{|L_{1,v}(1, \beta)|_v |L_{2,v}(1, \beta)|_v}{\max(1, |\beta|_v)^2} \\
&= \prod_{v \in S'} \frac{|L_{1,v}(1, \beta)|_v |L_{2,v}(1, \beta)|_v}{\max(1, |\beta|_v)^2} \prod_{v \in S \setminus S'} \frac{|L_{1,v}(1, \beta)|_v |L_{2,v}(1, \beta)|_v}{\max(1, |\beta|_v)^2} \\
&= \prod_{v \in S'} \frac{|1|_v |\alpha_v - \beta|_v}{\max(1, |\beta|_v)^2} \prod_{v \in S \setminus S'} \frac{|1|_v |\beta|_v}{\max(1, |\beta|_v)^2} \\
&\leq \prod_{v \in S'} |\alpha_v - \beta|_v \prod_{v \in S \setminus S'} 1 \\
&= \prod_{v \in S} \min(1, |\alpha_v - \beta|_v) \leq H(1, \beta)^{-2-\epsilon}
\end{aligned}$$

Then, all these points P are contained in a finite union of hyperplanes (these are just points). It shows that the numbers $\beta \in k$ which satisfy (7) and the additional condition (8) are at most finitely many. $\square$

2.3 Series expansion

An other important ingredient of our approach to the study of the equation $f(a^m, y) = b^n$, is constructing series convergent to relevant quantities. In fact, once we have an approximation, we can truncate it and then proceed with the tools of diophantine approximation. We first recall the implicit function theorem for complete fields (see [Ser92], Theorem 1, p.83).

Implicit function theorem 2.12. *Let k be a complete field with respect to a non-trivial absolute value, $f(X_1, \dots, X_n) \in k[X_1, \dots, X_n]$ be a polynomial and $P = (p_1, \dots, p_n)$ be a point of k^n . Suppose that $\frac{\partial f}{\partial X_1}(P) \neq 0$. Then there exists a power series $F \in k[[X_2, \dots, X_n]]$ and a radius $r > 0$ such that F is absolutely convergent in $B_{r, n-1}(0) \subseteq k^{n-1}$ and when $(x_1, \dots, x_n) \in B_{r, n}(P) \subseteq k^n$*

$$f(x_1, \dots, x_n) = 0 \Leftrightarrow y = F(x_2 - p_2, \dots, x_n - p_n)$$

We use the notation $B_{r, n}(P) = \{\mathbf{x} \in k^n \mid |\mathbf{x}| < r\}$ and $|(x_1, \dots, x_n)| = \max(|x_1|, \dots, |x_n|)$.

When we do not have the non-singularity condition on the partial derivative we can still obtain something using a Puiseux expansion. A formal Puiseux series is a formal power series with exponents rational numbers bounded from below and with bounded denominator

$$\alpha(X) = \sum_{j=j_0}^{\infty} c_j X^{j/e} \quad (9)$$

where $c_j \in k$ and e is a fixed integer. The field of (formal) Puiseux series is then given by

$$\bigcup_{e=1}^{\infty} k((X^{1/e}))$$

If k is a complete field there is a notion of convergence. The formal Puiseux series (9) is a convergent Puiseux series if $\sum_{j=0}^{\infty} c_j X^j$ converges absolutely in a neighbourhood of 0. They can be evaluated at a point $x \in k^*$ if $|x|$ is nonzero and small enough and we have chosen a n -th root of x . It was already known to Newton (and later rediscovered by Puiseux) that formal Puiseux series can be used to express the roots of a polynomial with coefficients in the ring of formal Laurent power series $k((X))$ (see the introduction of [Ron17] or [Wal78] IV§3 or for a modern discussion of this theorem).

Newton-Puiseux theorem 2.13. *Let k be a field of characteristic zero. The algebraic closure of $k((X))$ is*

$$\bigcup_{k'} \bigcup_{e=1}^{\infty} k'((X^{1/e}))$$

where the first union runs over all finite field extensions $k \rightarrow k'$. If k is also complete, the same holds with convergent Laurent series and convergent Puiseux series..

Example 2.14. Let k be a complete field of characteristic zero and $p(X, Y) \in k[X, Y]$ be a polynomial monic of degree d as a polynomial in Y . We can factor f as

$$f(X, Y) = (Y - \alpha_1(X)) \cdots (Y - \alpha_d(X))$$

where

$$\alpha_i(X) = \sum_{j=j_0}^{\infty} a_{i,j} X^{j/e}$$

are convergent Puiseux series, e is an integer and $a_{i,j}$ are all in a finite extension of k . This is an expansion around 0, but sometimes an expansion around infinity is useful. With a change of variable $X \rightarrow X^{-1}$ we can achieve a factorization

$$f(X, Y) = (Y - \alpha'_1(X)) \cdots (Y - \alpha'_d(X))$$

where

$$\alpha_i(X)' = \sum_{j=j'_0}^{\infty} a'_{i,j} X^{-j/e'}$$

converge at the points $x \in k$ with $|x|$ large enough. These series are sometimes called Puiseux series at infinity.

2.4 A lemma

This section contains a lemma already proved in [CZ00]. As for [CZ00], it is a main ingredient of the proof of Theorem 3.1 but it has some interest by itself. We start with some preliminary notions.

We denote with $\mathbb{G}_m$ the algebraic abelian group $A_1(\bar{\mathbb{Q}}) \setminus \{0\}$. As a group it is $\bar{\mathbb{Q}}^*$. The algebraic subgroups of $\mathbb{G}_m^n$ are precisely the sets

$$T_{a_1 \dots a_n} = \{\mathbf{x} \in \mathbb{G}_m^n \mid x_1^{a_1} \cdot \dots \cdot x_n^{a_n} = 1\}$$

for some integers $a_1, \dots, a_n$. Hence, the translates of algebraic subgroups of $\mathbb{G}_m^n$ are precisely the sets

$$\{\mathbf{x} \in \mathbb{G}_m^n \mid x_1^{a_1} \cdot \dots \cdot x_n^{a_n} = c\}$$

for some integers $a_1, \dots, a_n$ and some $c \in \bar{\mathbb{Q}}^*$. The following is a finiteness result regarding subgroups of $\mathbb{G}_m^n$, a special case of a theorem by Laurent [EE93].

Theorem 2.15. (Laurent). *Let G be a finitely generated subgroup of $(\bar{\mathbb{Q}}^*)^n$ and let Σ be any subset of G . Then the Zariski closure of Σ in $\mathbb{G}_m^n$ consists of a finite union of translates of algebraic subgroups of $\mathbb{G}_m^n$.*

With this theorem from Laurent we can easily prove a lemma of fundamental importance for the next sections. First we recall the concept of multiplicative independence.

Definition 2.16. Two numbers $a, b \in \bar{\mathbb{Q}}^*$ are *multiplicatively dependent* if there exist integers i, j not both zero such that $a^i b^j = 1$. Otherwise they are said to be *multiplicatively independent*.

Main Lemma 2.17. *Let $g \in \bar{\mathbb{Q}}[X, Y]$ be nonzero. Suppose that for some nonzero numbers $a, b \in \bar{\mathbb{Q}}^*$ not roots of unity the equation*

$$g(a^m, b^n) = 0$$

has infinitely many solutions $(m, n) \in \mathbb{Z} \times \mathbb{Z}$ with $\min(|m|, |n|) \rightarrow +\infty$, then there exist a nonzero $\eta \in \bar{\mathbb{Q}}^*$ and integers h, k , both nonzero such that $g(T^h, \eta T^k) = 0$. Moreover a and b are multiplicatively dependent. If m, n are always positive and $|a| < 1$ and $|b| < 1$ for some norm $|\cdot|$ of $\bar{\mathbb{Q}}$ then h and k can be chosen positive.

Proof. Suppose first that g does not depend on Y . Then we would have a nonzero polynomial with infinitely many zeros since a is not a root of unity. It cannot happen. Similarly, g does depend on X .

Let G be the finitely generated subgroup of $\mathbb{G}_m^2$ generated by the points $(a, 1)$ and $(1, b)$. Let Σ be the subset of $\mathbb{G}_m^2$ defined by the equation $g(x, y) = 0$. Laurent's theorem 2.15 tells us that the Zariski closure of $\Sigma \cap G$ is the union of finitely many translates of algebraic subgroups. This closure is of course contained in the algebraic set of dimension 1 Σ , then it must be the union of translates of algebraic subgroups of dimension 1 or 0. Note that the dimension of $g = 0$ is 1 because $g(X, Y)$ depends both on X and Y . Since a and b are not roots of unity their powers are all different, then the hypothesis implies that $\Sigma \cap G$ is infinite so it can't be covered only with finitely many translates of dimension 0. So there exists a one dimensional translate which contains infinitely many points of $\Sigma \cap G$. It is contained in the algebraic closure of $\Sigma \cap G$, which is contained in Σ . Such a translate is defined by an equation $x^k y^{-h} = c$ for some constants $h \in \mathbb{Z} \setminus \{0\}$, $k \in \mathbb{N} \setminus \{0\}$ and $c \in \bar{\mathbb{Q}}^*$. For some $\eta \in \bar{\mathbb{Q}}^*$ the point $(T^h, \eta T^k) \in \mathbb{G}_m^2$ is in this translate for every $T \in \bar{\mathbb{Q}}^*$. Therefore it is also in $g = 0$, that is

$$g(T^h, \eta T^k) = 0 \tag{10}$$

which is the first statement.

We have already discussed that an infinite subset of $\Sigma \cap G$ is contained in a translate $x^k y^{-h} = c$, since $\Sigma \cap G$ is covered by a finite number of them. This means that we have infinitely many pairs $(m, n) \in \mathbb{Z} \times \mathbb{Z}$ such that

$$a^{km} b^{-hn} = c \tag{11}$$

Consider two of these pairs $(m_1, n_1) \neq (m_2, n_2)$. Then $a^{k(m_1-m_2)} b^{-h(n_1-n_2)} = 1$ is a non trivial multiplicative relation between a and b .

Suppose now that m, n are positive and $|a|_v < 1$ and $|b|_v < 1$. From (11) we can write $|a|_v^{km} |b|_v^{-hn} = |c|_v$. Note that k, m, n are all positive integers, then also h has to be positive since the relation holds for infinitely many m, n with $\min(m, n) \rightarrow \infty$. $\square$

The same argument also proves the following extension to S -units.

Lemma 2.18. *Let $g \in \bar{\mathbb{Q}}[X, Y]$ be nonzero, $k \subset \bar{\mathbb{Q}}$ be a number field, and S be a finite set of places of k containing the archimedean ones. Suppose that there are infinitely many S -units x and y such that*

$$g(x, y) = 0$$

with infinitely many different values of x and of y . Then there exist a nonzero $\eta \in \bar{\mathbb{Q}}^$ and integers h, k , both not zero such that $g(T^h, \eta T^k) = 0$.*

3 The case of number fields

The following theorem is a natural generalization to number fields of the Main Theorem of [CZ00].

Theorem 3.1. *Let $f(X, Y) = a_0(X)Y^d + a_1(X)Y^{d-1} + \cdots + a_d(X)$ be a polynomial of degree $d \geq 2$ in Y with coefficients in a number field k . Let $a, b \in O_k \setminus \{0\}$ be two nonzero algebraic integers, not roots of unity. Suppose that*

(i) a_0 is constant

(ii) the equation $f(0, Y)$ has no repeated roots

(iii) there exists an absolute value $|\cdot|$ of k such that $|a| < 1$ and $|b| < 1$

If the equation

$$f(a^m, y) = b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$, then there exist an integer $h > 0$ and a polynomial $p(X) \in \bar{k}[X]$ such that $f(X^h, p(X))$ has only one term. Also a and b are multiplicatively dependent.

Remark 3.2. We assume a and b not roots of unity since otherwise the equation has infinitely many solutions as soon as it has one.

Remark 3.3. In the case $k = \mathbb{Q}$, (iii) is equivalent to a and b not coprime.

Remark 3.4. The coefficients of p may not be in k . Take for example the polynomial $f(X, Y) = Y + (X^3 - 2Y^2)(X^3 - 2Y^2 + 1)$. It satisfies the hypotheses of the theorem with $k = \mathbb{Q}$, $a = b = 2$. In fact, the equation $f(2^m, y) = 2^n$ has infinitely many solutions: if n and m are such that $3m = 2n + 1$ then

$$f(2^m, 2^n) = 2^n$$

There is a polynomial $p(X) = 2^{-1/2}X^3$ with the required property

$$f(X^2, 2^{-1/2}X^3) = 2^{-1/2}X^3$$

However there is no such polynomial with rational coefficients. We note that this fact was overlooked in the paper [CZ00]. We give a sketch of proof.

Proof. Suppose that $f(X^h, p(X)) = \eta X^k$ with $p(X) \in \mathbb{Q}[X]$ and $h > 0$. First, note that $p(X)$ is not 0 because $f(X^h, 0) = X^{3h}(X^{3h} + 1)$ is not a monomial. Let d and r be the degree and the zero-multiplicity in 0 of p . The key observation is that the degree and the zero-multiplicity in 0 of $X^{3h} - 2p(X)^2$ are respectively $\max(3h, 2d)$ and $\min(3h, 2r)$. Then, the degree of $f(X^h, p(X))$ is $2\max(3h, 2d)$ and $k = 2\max(3h, 2d) > 0$.

We evaluate at $X = 0$ and find that $p(0) - 2p(0)^2(1 - 2p(0)^2) = 0$ which implies $p(0) = 0$ since 0 is the only rational root of $X - 2X^2(1 - 2X^2) = 0$. Hence $r > 0$.

The zero-multiplicity in 0 of $f(X^h, p(X)) = p(X) + (X^{3h} - 2p(X)^2)(X^{3h} - 2p(X)^2 + 1)$ must be k . Note that $\min(3h, 2r)$ is the zero-multiplicity in 0 of $(X^{3h} - 2p(X)^2)(X^{3h} - 2p(X)^2 + 1)$. Then $r = \min(3h, 2r)$ for otherwise the zero-multiplicity in 0 of $f(X^h, p(X))$ would be $\min(3h, r) < k$.

So we have $r = 3h$ and $k = 2\max(3h, 2d) = 4d$. Also, the degree and the zero-multiplicity in 0 of $X^{3h} - 2p(X)^2$ are respectively $2d$ and r . Then, the leading term of $(X^3 - 2Y^2)(X^3 - 2Y^2 + 1)$ is cX^{4d} , the next (nonzero) higher order term has exponent at least $2d + r$. It should simplify with a term of the same order in p but it cannot happen because $2d + r > d$ since $r > 0$. $\square$

3.1 The proof

We start with a summary of the proof of Theorem 3.1. We subdivide three cases. The most interesting and general one is when $0 < \lim m/n < \infty$ that is $m \ll n \ll m$. By (ii) there is a completion of the number field k such that both a^m and b^n converge to 0. Hence y , on a subsequence, converges to a root of $f(0, Y) = 0$. Hypothesis (iii) is needed to use the implicit function theorem to write y as a series in a^m and b^n convergent when m and n are big enough. Next, we apply the subspace theorem to suitable points and we find that actually y must be given by a *fixed polynomial* in a^m and b^n for infinitely many solutions. Hence, we remain with only two unknowns: m and n . As we have already noted this is an easy situation. The last step is to apply Lemma 2.17 to this equation in two S -units unknown.

Proof. The equation $f(a^m, y) = b^n$ has an infinite sequence of solutions (m, n, y) with $\min(m, n) \rightarrow \infty$. The sequence m/n has a subsequence convergent to a value in $[0, +\infty]$. We consider now only the solutions in such a subsequence and we subdivide cases according to the value of the limit.

3.1.1 First case: $m/n \rightarrow 0$

Suppose that $|\cdot|$ is an absolute value of k such that $|b| > 1$. We will approximate with respect to this norm. First, note that $|y|$ is unbounded in any sequence where n tends to $+\infty$. In fact, suppose $|y|$ is bounded, then using that $m < \epsilon n$ for every $\epsilon > 0$ definitely, we see that $|f(a^m, y)| \ll |a|^{Cm} \ll |b|^{\epsilon n}$, for some $C > 0$, while $|f(a^m, y)| = |b|^n$. In particular $|y| > 1$ definitely.

Let $\Delta \in \mathbb{N}$ be a common denominator for the coefficients of $a_1(X)/(a_0d)$. That is for every coefficient c_i there is $t \in O_k$ such that $c_i = t/\Delta$. We can suppose $a_1(X) = 0$. In fact, we can change variables replacing Y with $Y/\Delta - a_1(X)/(a_0d)$ to obtain a new polynomial $f_1(X, Y) = a'_0Y^d + a'_2(X)Y^{d-2} + \dots + a'_d(X)$. The hypotheses remain satisfied for the new equation: an integral solution (m, n, y) to the original equation determines an integral solution $(m, n, \Delta(y + a_1(X)/(a_0d)))$ for the new equation. Also the conclusion for the new equation implies the conclusion for the original equation: suppose that $f_1(X^h, p(X)) = \eta X^k$ then $f(X^h, p(X)/\Delta - a_1(X^h)/(a_0d)) = \eta X^k$ so that $f(X^h, p'(X)) = \eta X^k$ for some polynomial p' .

Choose an integer r such that for an infinite subsequence of solutions we have $n = r + ld$ for some integer l . We divide both sides of the equation $f(a^m, y) = b^n$ by $b^r y^d$ to get

$$\left(\frac{b^l}{y}\right)^d - \frac{a_0}{b^r} = b^{-r}[a_2(a^m)y^{-2} + \dots + a_d(a^m)y^{-d}] \quad (12)$$

Observe that $m/l \rightarrow 0$. Recall that $|y| > 1$ definitely and $|b| > 1$, moreover $|a^m| \ll |b|^{\epsilon l}$ for every $\epsilon > 0$, then

$$|b^{-r}[a_2(a^m)y^{-2} + \dots + a_d(a^m)y^{-d}]| \ll |b|^{\epsilon l} \quad (13)$$

and also the same quantity multiplied by y^2

$$|b^{-r}[a_2(a^m) + \dots + a_d(a^m)y^{2-d}]| \ll |b|^{\epsilon l} \quad (14)$$

Combining (12) and (13) we get

$$\left|\frac{b^l}{y}\right|^d \ll |b|^{\epsilon l}$$

implying $|y| \gg |b|^{l-\epsilon'}$ for every $\epsilon' > 0$. Then using (14) we find from (12) that for every $\delta > 0$

$$\left|\left(\frac{b^l}{y}\right)^d - \frac{a_0}{b^r}\right| \ll |y|^{-2}|b|^{\epsilon l} \ll |y|^{-2+\delta} \quad (15)$$

Note that a_0/b^r is constant and let $\alpha_1, \dots, \alpha_d$ be its d -th roots. We have the factorization

$$\left(\frac{b^l}{y}\right)^d - \frac{a_0}{b^r} = \prod_{i=1}^d \left(\left(\frac{b^l}{y}\right) - \alpha_i\right)$$

Also, all except at most one of the factors of the above product are bigger than $\min_{i \neq j} |\alpha_i - \alpha_j|/2$ so that

$$\min_i \left| \left(\frac{b^l}{y}\right) - \alpha_i \right| \ll \left| \left(\frac{b^l}{y}\right)^d - \frac{a_0}{b^r} \right|$$

Then from (15) we find the existence of an i (possibly different for each solution) such that

$$\left| \alpha_i - \frac{b^l}{y} \right| \ll |y|^{-2+\delta} \quad (16)$$

We will suppose i to be constant by passing to a subsequence.

Let S' be the finite set of places v for which $|b|_v > 1$. For each of them (16) holds for an appropriate α_i . Note that S' is not empty by Theorem 2.2 and 2.3 because b is not a root of unity. We have obtained an infinite sequence of solutions (m, n, y) such that for every $v \in S'$ there exists an algebraic number $\alpha_v \in \bar{\mathbb{Q}}$ such that putting $n = r + ld$ we have

$$\left| \alpha_v - \frac{b^l}{y} \right|_v \ll |y|_v^{-2+\delta} \quad \text{for every } v \in S'$$

We have already noted that $|y|_v > 1$, then the previous inequality implies that $|b^l|_v$ is asymptotic to $|\alpha_v|_v \cdot |y|_v$. With this in mind we can also write

$$\left| \alpha_v - \frac{b^l}{y} \right|_v \ll |b^l|_v^{-2+\delta} \quad \text{for every } v \in S' \quad (17)$$

We now apply the Generalized Roth's Theorem 2.11 with S containing all the archimedean places and those dividing b so that b is an S -unit. We choose the previous α_v when v is in S' . For the other absolute values v not in S' we choose $\alpha_v = 0$. Then, for every $\epsilon > 0$ there exist at most finitely many numbers $\beta \in k$ such that

$$\prod_{v \in S} \min(1, |\alpha_v - \beta|_v) \ll H(\beta)^{-2-\epsilon} \quad (18)$$

We proceed to show that (18) holds for some $\epsilon > 0$ when $\beta = b^l/y$. Since b^l is an S -unit and y is an integer we have

$$H(\beta) = H(\beta^{-1}) = \prod_{v \in M_k} \max\left(1, \left|\frac{y}{b^l}\right|_v\right) = \prod_{v \in S} \max\left(1, \left|\frac{y}{b^l}\right|_v\right)$$

Moreover $\prod_{v \in S'} \max(1, |b^l/y|_v)$ is bounded since $|b^l/y|_v$ is close to $|\alpha_v|_v$ up to a bounded term by (17). With this estimates on $H(\beta)$, to get (18) it is enough to prove

$$\prod_{v \in S} \min(1, |\alpha_v - \beta|_v) \ll \prod_{v \in S \setminus S'} \max\left(1, \left|\frac{y}{b^l}\right|_v\right)^{-2-\epsilon}$$

Thanks to (17) it is sufficient

$$\prod_{v \in S'} |b^l|_v^{-2+\delta} \prod_{v \in S \setminus S'} \min\left(1, \left|\frac{b^l}{y}\right|_v\right) \ll \prod_{v \in S \setminus S'} \max\left(1, \left|\frac{y}{b^l}\right|_v\right)^{-2-\epsilon}$$

which is

$$\prod_{v \in S'} |b^l|_v^{-2+\delta} \ll \prod_{v \in S \setminus S'} \max\left(1, \left|\frac{y}{b^l}\right|_v\right)^{-1-\epsilon} \quad (19)$$

Observe that for v in $S \setminus S'$ we know $|b|_v \leq 1$, then $|y|_v$ is bounded, for every $\mu > 0$, by a constant multiple of $\exp(\mu l)$ since $|a|_v, |b|_v \ll \exp(\mu l)$ and y satisfies $f(a^m, y) = b^n$. Hence

$$\max\left(1, \left|\frac{y}{b^l}\right|_v\right) \ll \max\left(1, \left|\frac{1}{b^l}\right|_v\right) \exp(\mu l) = \left|\frac{1}{b^l}\right|_v \exp(\mu l) \quad \text{for every } v \in S \setminus S'$$

Continuing from (19), we just need

$$\prod_{v \in S'} |b^l|_v^{-2+\delta} \ll \prod_{v \in S \setminus S'} \left(\left|\frac{1}{b^l}\right|_v \exp(\mu l)\right)^{-1-\epsilon} \quad (20)$$

For small $\mu > 0$ we have $|b^l|_v > \exp(\mu l)$ for every $v \in S'$ just because $|b|_v > 1$. Then, the following is enough (we can modify $\delta > 0$ as needed so that inequality (21) below implies the desired inequality (20))

$$\prod_{v \in S'} |b^l|_v^{-2+\delta} \ll \prod_{v \in S \setminus S'} \left(\left|\frac{1}{b^l}\right|_v\right)^{-1-\epsilon} \quad (21)$$

Put $\epsilon = \delta = \frac{1}{2}$, then we can establish (21) (hence, the wanted inequality (18)) since

$$\prod_{v \in S'} |b^l|_v^{-3/2} \ll \prod_{v \in S \setminus S'} |b^l|_v^{3/2}$$

is actually an equality thanks to the Product Formula 2.3 because b is an S -unit.

From the generalized Roth's theorem it follows that the quantity $\beta = b^l/y$ takes only finitely many values, therefore $y = cb^l$ holds for some constant c and for infinitely many solutions. Consider the polynomial $g(X, Y) := f(X, cY) - b^r Y^d$. We get infinitely many solutions (m, l) for the equation $g(a^m, b^l) = 0$. If $g = 0$ identically, then $f(X, cY)$ would be a monomial in Y of degree ≥ 2 , contrary to assumption (ii). We apply Lemma 2.17 to find that there exist positive integers h, k (here we use hypothesis (iii)) and a number $\eta \in \mathbb{Q}^*$ such that

$$g(T^h, \eta T^k) = 0$$

that is

$$f(T^h, c\eta T^k) = b^r T^{kd} \quad (22)$$

which is enough to conclude. Moreover a and b are multiplicatively dependent by the same Lemma 2.17.

Remark 3.5. For this case we have barely used hypothesis (iii). Without it, the only difference is that the k appearing in (22) is not guaranteed to be positive (we need (iii) when applying Lemma 2.17 to get h, k both positive). So without (iii) we only find the existence of a rational function $p(X) \in \bar{k}[X, X^{-1}]$ (not necessarily a polynomial) such that $f(X^h, p(X))$ has only one term. Also hypothesis (ii) could be changed with the much weaker condition that $f(X, Y)$ is not a monomial in Y .

3.1.2 Second case: $0 < \lim m/n < \infty$

Put $x = a^m$ and $z = b^n$. Let w be an absolute value such that $|a|_w < 1$ and $|b|_w < 1$, it exists by hypothesis (iii). Consider the completion k_w of k with respect to w . Then a^m and b^n converge to 0. Passing to a subsequence we can assume that y converges to a number $y_0 \in k_w$ (since k_w is locally compact and $|y|_w$ is bounded because $|a|_w < 1$ and $|b|_w < 1$) which must satisfy $f(0, y_0) = 0$ by a continuity argument. We now use the implicit function Theorem 2.12 over k_w . We consider the surface $f(x, y) = z$ and we write y as a function of x and z in a neighborhood of the point $(x, y, z) = (0, y_0, 0)$. Thanks to hypothesis (ii), y admits a series expansion converging for (x, z) in a neighborhood of $(0, 0)$ given by

$$y = \sum_{i,j \geq 0} c_{i,j} x^i z^j \quad (23)$$

where $c_{i,j}$ are numbers in k_v . Since $f \in k[X, Y]$ the $c_{i,j}$ are algebraic over k so that we may think as if $c_{i,j} \in \bar{\mathbb{Q}}$. When $x = a^m$ and $z = b^n$ the series (23) converges to y , definitely in n .

Fix a real number $H > 0$, define

$$\mathcal{A}_H := \{(i, j) : i \geq 0, j \geq 0, i + j \leq H\}$$

and put

$$S(m, n) = \sum_{(i,j) \in \mathcal{A}_H} c_{i,j} a^{mi} b^{nj}$$

Using that $n \ll m \ll n$ we obtain from (23), for the sequence of solution under consideration, the inequality

$$|y - S(m, n)|_w < \theta^{nH} \quad (24)$$

for large n , where $\theta < 1$ is a constant. This inequality regarding the truncation of a convergent series derives from the following lemma.

Lemma 3.6. *Let k be a complete field with respect to the norm $|\cdot|$, and let $F \in k[[X_1, \dots, X_n]]$ be a power series*

$$F(X_1, \dots, X_n) = \sum_{i_1 \geq 0, \dots, i_n \geq 0} a_{i_1, \dots, i_n} X_1^{i_1} \cdots X_n^{i_n}$$

absolutely convergent in a neighbourhood of 0 with $a_{i_1, \dots, i_n} \in k$. Suppose that it converges at the points $\mathbf{x} \in k^n$ such that $|\mathbf{x}| < r$ for some given $r > 0$. We recall that $|(x_1, \dots, x_n)| = \max(|x_1|, \dots, |x_n|)$. Fix an integer $H > 0$ and let F_H be the series F truncated to the order H , that is the polynomial made of the terms in F of total degree at most H . Then, for every $\mathbf{x} \in k^n$ such that $|\mathbf{x}| < r/2$ we have the bound

$$|F(\mathbf{x}) - F_H(\mathbf{x})| \leq C \left(\frac{2|\mathbf{x}|}{r} \right)^H$$

for some constant C depending only on F .

Proof. Put $A_k = \sum_{i_1 + \dots + i_n = k} |a_{i_1, \dots, i_n}|$. Since $F(\mathbf{x})$ absolutely converges when $|\mathbf{x}| < r$, from the theory of the radius of convergence of a one variable series we get that

$\limsup_{k \rightarrow \infty} A_k^{1/k} \leq r^{-1}$. Then

$$\begin{aligned}
|F(\mathbf{x}) - F_H(\mathbf{x})| &= \left| \sum_{i_1 + \dots + i_n \geq H} a_{i_1, \dots, i_n} x_1^{i_1} \cdots x_n^{i_n} \right| \\
&\leq \sum_{k \geq H} \left| \sum_{i_1 + \dots + i_n = k} a_{i_1, \dots, i_n} x_1^{i_1} \cdots x_n^{i_n} \right| \\
&\leq \sum_{k \geq H} A_k |\mathbf{x}|^k \\
&= \sum_{k \geq H} A_k \left(\frac{2|\mathbf{x}|}{r} \right)^k \left(\frac{r}{2} \right)^k \\
&\leq \left(\frac{2|\mathbf{x}|}{r} \right)^H \sum_{k \geq H} A_k \left(\frac{r}{2} \right)^k \\
&\leq \left(\frac{2|\mathbf{x}|}{r} \right)^H \sum_{k \geq 0} A_k \left(\frac{r}{2} \right)^k = C \left(\frac{2|\mathbf{x}|}{r} \right)^H
\end{aligned}$$

where $C = \sum_{k \geq 0} A_k \left(\frac{r}{2} \right)^k$ is a constant, not infinite since F absolutely converges at the points $\mathbf{x} \in \bar{\mathbf{k}}^n$ such that $|\mathbf{x}| < r$. $\square$

Let $\mathcal{B}_H$ be a minimal subset of $\mathcal{A}_H$ with the following property: there exist numbers $\lambda_{i,j}$ in $\mathbf{k}_w$ algebraic over $\mathbf{k}$ for $(i, j) \in \mathcal{B}_H$ such that for an infinite subsequence of the solutions we are considering we have

$$S(m, n) = \sum_{(i,j) \in \mathcal{B}_H} \lambda_{i,j} a^{mi} b^{nj} \quad (25)$$

Observe that $\mathcal{B}_H$ exists since $\mathcal{A}_H$ has the stated property.

Let $\#\mathcal{B}_H = N$ and fix an enumeration of the N pairs in $\mathcal{B}_H$. For $h = 1, \dots, N$ put $\beta_h = \beta_h(n, m) = a^{mi} b^{nj}$ if (i, j) is the h -th pair of $\mathcal{B}_H$. We have from (24) and (25)

$$|y - \lambda_1 \beta_1 - \dots - \lambda_N \beta_N|_v < \theta^{nH} \quad (26)$$

for suitable $\lambda_i \in \mathbf{k}_w$ algebraic over $\mathbf{k}$. We apply the Subspace Theorem 2.8 with S a finite set of places containing the archimedean ones such that a and b are S -units. For a vector $T = (T_0, \dots, T_N) \in \mathbf{k}^{N+1}$ we define linear forms

$$\begin{aligned}
L_{v,i}(T) &= T_i \quad \text{if } v \neq w \text{ or } i \neq 0 \\
\text{and } L_{w,0}(T) &= T_0 - \lambda_1 T_1 - \dots - \lambda_N T_N
\end{aligned}$$

These linear forms are linearly independent for every v and the coefficients are algebraic over k . For a solution (m, n, y) let $P = (y, \beta_1, \dots, \beta_N)$ be a point in O_S^{N+1} with all S -integer coordinates. Let us estimate

$$\begin{aligned} \prod_{v \in S} \prod_{i=0}^N |L_{v,i}(P)|_v &= |L_{w,0}(P)|_w \left(\prod_{v \in S \setminus \{w\}} |y|_v \right) \left(\prod_{i=1}^N \prod_{v \in S} |\beta_i|_v \right) \\ &= |L_{w,0}(P)|_w \prod_{v \in S \setminus \{w\}} |y|_v \quad \text{since the } \beta_i \text{ are } S\text{-units} \end{aligned}$$

If it happens (for an H large enough and an $\epsilon > 0$ small enough) that for infinitely many solutions

$$|L_{w,0}(P)|_w \prod_{v \in S \setminus \{w\}} |y|_v \ll H(P)^{-\epsilon} \quad (27)$$

then applying the Subspace Theorem 2.8 we would find an hyperplane which contains infinitely many points $P = (y, \beta_1, \dots, \beta_N)$. It gives a non trivial linear relation with $A_i \in k$

$$A_0 y + A_1 \beta_1 + \dots + A_N \beta_N = 0 \quad (28)$$

holding for an infinite subsequence of solutions.

To find H and ϵ such that (27) holds we just need the following three bounds

$$\begin{aligned} |L_{w,0}(P)|_w &< \theta^{nH} \quad \text{by (26)} \\ H(P) &\ll C^{nHD} \end{aligned} \quad (29)$$

$$\prod_{v \in S \setminus \{w\}} |y|_v \ll C^{nD} \quad (30)$$

for some $C > 1$ and $D > 0$. We now prove these bounds.

We know that $n \ll m \ll n$, hence there exists $C > 1$ such that $|x|_v, |z|_v \ll C^n$ for every absolute value v . Then $|y|_v$ is also bounded by C^n because it satisfies $f(x, y) = z$. Moreover, the $|\beta_i|_v$ are bounded by C^{nH} for every absolute value v because they are monomials in x, z of total degree at most H . Hence, if we put D

the cardinality of S we have bound (29)

$$\begin{aligned}
H(P) &= H(y, \beta_1, \dots, \beta_N) \\
&= \prod_{v \in M_k} \max(|y|_v, |\beta_1|_v, \dots, |\beta_N|_v) \\
&\leq \prod_{v \in S} \max(|y|_v, |\beta_1|_v, \dots, |\beta_N|_v) \quad \text{because } y \text{ and the } \beta_i \text{ are } S\text{-integers} \\
&\ll C^{mHD}
\end{aligned}$$

and we also have bound (30)

$$\prod_{v \in S \setminus \{w\}} |y|_v \ll C^{mD}$$

To obtain (27) it is therefore sufficient that

$$C^{Dn(H\epsilon+1)} \theta^{nH} \ll 1$$

which, for a suitable choice of a large H and a small ϵ , can clearly be guaranteed to hold for every n . We deduce the existence of a linear relation of the form (28).

Suppose first that $A_0 = 0$. Then some other coefficient must be non-zero, say A_1 . We can express β_1 as a linear combination of the other β_i and we can rewrite $S(m, n) = \sum_{(i,j) \in \mathcal{B}_H} \lambda_{i,j} a^{mi} b^{nj}$ without using β_1 , contradicting the minimality of $\mathcal{B}_H$.

Then $A_0 \neq 0$ and

$$y = - \left(\frac{A_1}{A_0} \beta_1 + \dots + \frac{A_N}{A_0} \beta_N \right)$$

hence, y is a fixed polynomial in a^m and b^n for an infinite sequence of solution, that is there exists a polynomial $q(X, Z) \in k[X, Z]$ such that $y = q(a^m, b^n)$. If we define the polynomial $g(X, Z) := f(X, q(X, Z)) - Z$ then the equation $g(a^m, b^n) = 0$ has infinitely many solutions (m, n) . Observe that g is not identically zero because if $\deg_Z q \geq 1$ then $\deg_Z f(X, q(X, Z)) \geq d \geq 2$, if instead $\deg_Z q = 0$ then $\deg_Z g = 1$. Again, Lemma 2.17 applied to g gives the wanted statement. There exist $\eta \in \bar{\mathbb{Q}}$ and integers $h > 0$ and $k > 0$ such that

$$0 = g(X^h, \eta X^k) = f(X^h, q(X^h, \eta X^k)) - \eta X^k$$

and we can put $p(X) = q(X^h, \eta X^k)$ so that $f(X^h, p(X))$ has only one term. Moreover a and b are multiplicatively dependent by the same Lemma 2.17.

3.1.3 Third case: $m/n \rightarrow \infty$

Let $|\cdot|$ be an absolute value of k . We can factor $f(X, Y)$, viewed as a polynomial in Y , in the algebraic closure of $k(X)$

$$f(X, Y) = a_0(Y - \alpha_1(X)) \cdots (Y - \alpha_d(X)) \quad (31)$$

where the functions $\alpha_1 \dots \alpha_d$ are Puiseux series at infinity of the form

$$\alpha_i(X) = \sum_{j=-h}^{\infty} a_{i,j} X^{-j/e}$$

convergent at the points $x \in k$ such that $|x|$ is large enough (see Example 2.14). The $a_{i,j}$ should be chosen in a finite extension of the completion of k with respect to $|\cdot|$. Since $f \in k(X, Y)$ the $a_{i,j}$ are algebraic over k so we may think as if $a_{i,j} \in \bar{\mathbb{Q}}$.

We reduce to the case $e = 1$ in a standard way. Consider r such that $n \equiv r \pmod{e}$ for infinitely many solutions. Replacing X by $a^r X^e$ we obtain a new equation $f_1(a^m, y) = b^n$ with infinitely many solution but this time $f_1(X, Y)$ has roots $\alpha_i(a^r X^e)$ which are Laurent series in X^{-1} . A solution (m, n, y) to $f(a^m, y) = b^n$ determines a solution $((m - r)/e, n, y)$ of the new equation. If there exist integers $h, k > 0$, a number $\eta \in \bar{k}$ and a polynomial $p(X)$ such that $f_1(X^h, p(X)) = \eta X^k$ then $f(a^r X^{eh}, p(X)) = \eta X^k$. This can also be rewritten as $f(X^{eh}, p(X/a^{r/eh})) = \eta/a^{kr/eh} X^k$ which is the wanted statement for f . We can therefore suppose $e = 1$.

For a solution (m, n, y) write $x = a^m$ and $z = b^n$. Suppose that $|\cdot|$ is an absolute value such that $|a| > 1$. Since $m/n \rightarrow \infty$, for every positive δ we have

$$|f(a^m, y)| = |b|^n \ll |a|^{\delta m} = |x|^\delta$$

which is

$$|y - \alpha_1(x)| \cdots |y - \alpha_d(x)| \ll |x|^\delta \quad (32)$$

Let k be the minimal index such that the coefficients $a_{1,k}, \dots, a_{d,k}$ are not all equal. We remark at once that $k \leq 0$. In fact, assume the contrary and put $P(X) = \sum_{j=-h}^0 a_{1,j} X^{-j} \in k[X]$. Then we can factor $f_1(X, Y) := f(X, Y - P(X))$ as

$$f_1(X, Y) = a_0(Y - \alpha'_1(X)) \cdots (Y - \alpha'_d(X)) \quad (33)$$

where $\alpha'_i(X) = \alpha_i(X) - P(X)$ are Puiseux series given explicitly by

$$\alpha'_i(X) = \sum_{j=1}^{\infty} a_{i,j} X^{-j/e}$$

Note that now all the exponents of $\alpha'_i(X)$ are negative, therefore from (33) we see that all the coefficients of Y^j in f_1 , for $j < d$, must be zero, for they are polynomials vanishing at ∞ . So f_1 and hence f would be a perfect d -th power in $\mathbb{Q}[X, Y]$, contrary to assumption (ii). We have proved $k \leq 0$.

Since for each j the set $\{a_{1,j}, \dots, a_{d,j}\}$ is invariant under conjugation over k , the algebraic numbers $a_{i,j}$ with $j < k$ are all in k . Let $\Delta \in \mathbb{N}$ be a common denominator of the elements $a_{1,-h}, \dots, a_{1,k-1}$. We can suppose $k = -h$ just by substituting Y with $\frac{Y}{\Delta} + \sum_{j=-h}^{k-1} a_{1,j} X^{-j}$ to get a new polynomial f_1 . A solution (m, n, y) to $f(a^m, y) = b^n$ determines a solution with $y' = \Delta(y - \sum_{j=-h}^{k-1} a_{1,j} a^{-mj})$ of the equation $f_1(a^m, y') = b^n$. Note that y' is an integer since $k - 1 < 0$. Moreover, if $f_1(X^h, p(X)) = \eta X^k$ then $f(X^h, p(X))/\Delta + \sum_{j=-h}^{k-1} a_{1,j} X^{-j} = \eta X^k$ so that we have the wanted statement for f . Note, again, that $\sum_{j=-h}^{k-1} a_{1,j} X^{-j}$ is a polynomial since $k - 1 < 0$. The roots of $f_1(X, Y)$, viewed as a polynomial in Y are given by the Puiseux series $\Delta \left(\alpha_i(X) - \sum_{j=-h}^{k-1} a_{1,j} X^{-j} \right)$. We therefore assume $k = -h$, that is the coefficients of X^h in $\alpha_1, \dots, \alpha_d$ are not all equal.

Suppose now that for an infinite subsequence of solutions $|x| = |a|^m$ is so large that all the series α_i converge at x . Suppose also, as we may, that

$$|\alpha_1(x) - y| = \min_i (|\alpha_i(x) - y|) \quad (34)$$

Renumbering the indexes we can write $a_{1,-h} = \dots = a_{t,-h}$ and $a_{i,-h} \neq a_{1,-h}$ for $t < i \leq d$ for an integer $t < d$. We have

$$|y - \alpha_i(x)| \geq |\alpha_1(x) - \alpha_i(x)| - |y - \alpha_1(x)| \geq |\alpha_1(x) - \alpha_i(x)| - |y - \alpha_i(x)|$$

whence

$$|y - \alpha_i(x)| \geq \frac{|\alpha_1(x) - \alpha_i(x)|}{2} \quad (35)$$

Observe that $|\alpha_1(x) - \alpha_i(x)| \gg |x|^h$ when $i > t$. Then (35) implies

$$|y - \alpha_i(x)| \gg |x|^h \quad \text{for } i > t \quad (36)$$

From (32) and (36) we then get

$$|y - \alpha_1(x)| \cdots |y - \alpha_t(x)| \ll |x|^{\delta - h(d-t)}$$

and using (34)

$$\begin{aligned} |y - \alpha_1(x)|^t &\ll |x|^{\delta - h(d-t)} \\ \text{equivalently } |y - \alpha_1(x)| &\ll |x|^{(\delta - h(d-t))/t} \end{aligned} \quad (37)$$

Assume first that $h \leq 0$. Then in the factorization (31) the $\alpha_i(X)$ are functions bounded at infinity when $|X| \rightarrow \infty$. Hence, the coefficients of $f(X, Y)$, viewed as polynomials in Y , are polynomials in X which are bounded at ∞ . Therefore they are all constant and $f(X, Y) = f(Y)$ in fact does not depend on X . In particular, if (m, n, y) is a solution of the equation then (m', n, y) is also a solution, for all $m' \in \mathbb{Z}$. In this case we may conclude the proof by putting either $m' = \lfloor \sqrt{n} \rfloor$, falling in the first case, or $m' = n$, falling in the second case, both cases being treated above. From now on we shall assume $h > 0$.

Let N be an integer such that $N - h > h(d - t)/t$. We use Lemma 3.6 (applied after a change of variables $X \rightarrow X^{-1}$) to get an estimate for the truncation of the series $\alpha_1(X)$ holding for some constant $r > 0$ when $|x|$ is big enough

$$\left| \alpha_1(x) - \sum_{j=-h}^{N-h-1} a_{1,j} x^{-j} \right| \ll \left(\frac{r|x|}{2} \right)^{-(N-h)}$$

Then, from this inequality and from (37) we get

$$\left| y - \sum_{j=-h}^{N-h-1} a_{1,j} x^{-j} \right| \ll |x|^{(\delta-h(d-t))/t} + \left(\frac{r|x|}{2} \right)^{-(N-h)} \ll |x|^{(\delta-h(d-t))/t}$$

because $-(N - h) < \delta - h(d - t)/t$ for every $\delta > 0$ and $|x| > 1$ definitely. This inequality can be obtained for every absolute value such that $|a| > 1$ however we remark that the number t and the index minimizing (34) depend on the absolute value chosen. With this in mind we affirm that for every v such that $|a|_v > 1$ there exist numbers t_v with $0 < t_v < d$ and $1 \leq i(v) \leq d$ such that for every N large enough we have

$$\left| y - \sum_{j=-h}^{N-h-1} a_{i(v),j} x^{-j} \right|_v \ll |x|_v^{(\delta-h(d-t_v))/t_v} \quad (38)$$

holding for each solution we are considering and for every $\delta > 0$.

Let S be the set containing all the archimedean absolute values and those dividing a so that a is an S -unit. Moreover, let $S' \subseteq S$ be the set of all the places v such that $|a|_v > 1$. We choose linear forms $L_{v,j}(T_0, \dots, T_N)$ for $j = 0, \dots, N$ and $v \in S$ defined as

$$\begin{aligned} L_{v,j}(T) &= T_j && \text{if } v \in S \setminus S' \text{ or } j \neq 0 \\ \text{and } L_{v,0}(T) &= T_0 - \sum_{j=-h}^{N-h-1} a_{i(v),j} T_{j+h+1} && \text{if } v \in S' \end{aligned}$$

For a solution (m, n, y) put $P = (y, x^h, \dots, x^{-(N-h-1)}) \in O_S^{N+1}$ and consider the double product

$$\begin{aligned} \prod_{v \in S} \prod_{i=0}^N |L_{v,i}(P)|_v &= \left(\prod_{v \in S'} |L_{v,0}(P)|_v \right) \left(\prod_{v \in S \setminus S'} |y|_v \right) \left(\prod_{j=-h}^{N-h-1} \prod_{v \in S} |x^{-j}|_v \right) \\ &= \left(\prod_{v \in S'} |L_{v,0}(P)|_v \right) \left(\prod_{v \in S \setminus S'} |y|_v \right) \quad \text{because } x \in O_S^* \end{aligned}$$

We would like to find $\epsilon > 0$ small enough such that

$$\prod_{v \in S} \prod_{i=0}^N |L_{v,i}(P)|_v = \left(\prod_{v \in S'} |L_{v,0}(P)|_v \right) \left(\prod_{v \in S \setminus S'} |y|_v \right) \ll H(P)^{-\epsilon} \quad (39)$$

We proceed to estimate each piece appearing in the previous equation. Since $m \gg n$ there exists a constant $C > 1$ such that $C^{-m} < |x|, |z| < C^m$ for any norm. Then, also $|y| \ll C^m$ because y satisfies $f(x, y) = z$. Hence, if D is the cardinality of S we have

$$\begin{aligned} H(P) &= H(y, x^h, \dots, x^{-(N-h-1)}) \\ &= \prod_{v \in M_k} \max(|y|_v, |x^h|_v, \dots, |x^{-(N-h-1)}|_v) \\ &\leq \prod_{v \in S} \max(|y|_v, |x^h|_v, \dots, |x^{-(N-h-1)}|_v) \quad \text{because } y, x \in O_S \\ &\ll C^{mD \max(|h|, |N-h-1|)} \ll C^{mDN} \quad \text{if } N \text{ is large} \end{aligned} \quad (40)$$

If $v \in S \setminus S'$ then for every $\mu > 0$, $|a| \leq 1$ and $|b| < C^{m\mu}$ because $\lim m/n = \infty$, hence

$$|y|_v \ll C^{m\mu} \quad \text{for } v \in S \setminus S' \quad (41)$$

since y satisfies $f(a^m, y) = b^n$.

Now we choose $\delta > 0$ so small that $(\delta - h(d - t_v))/t_v < 0$ for every $v \in S'$ (recall that $h > 0$ and $0 < t_v < d$). Let $r > 0$ be such that $|a|_v > C^r$ for every $v \in S'$. It exists since both $|a|_v$ and C are greater than 1 and S' is finite. We have

$$|x|_v > C^{mr} \quad (42)$$

Recall that by (38) and the definition of L_{v_0}

$$|L_{v,0}(P)|_v \leq \prod_{v \in S'} |x|_v^{(\delta - h(d - t_v))/t_v} \quad \text{for } v \in S'$$

Put $s := r \sum_{v \in S'} (\delta - h(d - t_v))/t_v$ a negative real number. Then, using (42), we finally have

$$\prod_{v \in S'} |L_{v,0}(P)|_v \leq \prod_{v \in S'} |x|_v^{(\delta - h(d - t_v))/t_v} \ll C^{ms} \quad (43)$$

With the estimates (40), (41) and (43), we get (39) if

$$C^{ms} C^{m\mu D} C^{mDN\epsilon} \ll 1$$

that is (taking the m -th root)

$$C^{s+\mu D+DN\epsilon} \ll 1$$

Since $s < 0$ we can choose μ and ϵ so that the previous inequality holds. Therefore we have established (39) and the Subspace Theorem 2.8 applies giving a linear relation holding for an infinite subsequence of solutions

$$A_0 y + A_1 x^h + \cdots + A_N x^{-N+h+1} = 0$$

Now, A_0 cannot vanish otherwise we would find infinitely many roots for the nonzero rational function $A_1 x^h + \cdots + A_N x^{-N+h+1} = 0$, in fact $x = a^m$ are distinct for distinct m because a is not a root of unity.

Hence

$$y = - \left(\frac{A_1}{A_0} x^h + \cdots + \frac{A_N}{A_0} x^{-N+h+1} \right) \quad (44)$$

We now show that $A_i = 0$ if $h - i + 1 < 0$ so that the right hand side of the above equation is a polynomial in x . Recall that thanks to hypothesis (iii) we have a place w such that $|b|_w, |a|_w < 1$. Let t be the biggest index such that $A_t \neq 0$. When $|x|$ is small enough from (44) we get

$$|y|_w \gg |x|_w^{h-t+1}$$

If, by contradiction, $h - t + 1 < 0$, then $|y|_w > C^m$ for some $C > 1$ but $|y|_w$ must be bounded since it satisfies $f(a^m, y) = b^n$ with f a monic polynomial in y and $|b|_w, |a|_w < 1$.

Therefore the right hand side of (44) is a polynomial in x meaning that there exists a polynomial $p \in k[X]$ such that $y = p(x)$ for the solutions we are considering. Put $g(X, Z) = f(X, p(X)) - Z$. As before, g is not zero and Lemma 2.17 applied to g gives the wanted statement. This is the end of the proof of Theorem 3.1. $\square$

Remark 3.7. For this third case we have barely used hypothesis (iii). Without it we can only say the conclusion holds for some rational function $p(x) \in \bar{k}[X, X^{-1}]$, not necessarily a polynomial. In fact (iii) was needed only to show that the right hand side of (44) is a polynomial in x , and to find that the h and k coming from Lemma 2.17 are both positive. Also hypothesis (ii) could be changed with the much weaker condition that $f(X, Y)$ is not a d -th perfect power in $\bar{\mathbb{Q}}[X, Y]$.

Remark 3.8. Under the hypotheses of Theorem 3.1 for all but finitely many solutions we have that

$$y = p(a^{m/h})$$

for some $h > 0$ and some polynomial $p(X)$ such that $f(X^h, p(X))$ is a monomial. This holds for a suitable choice of a root of a . We sketch how to prove this fact.

Proof. If we put $x = a^m$, $y = b^n$, then for all but finitely many solutions $y = F(x, z)$ and F can be chosen in a finite set of polynomials (this is a consequence of the subspace theorem or the generalized Roth's theorem in all the three cases of the above proof). Inspection of the proof of Lemma 2.17 also shows that for all but finitely many solutions we have $x^k = cz^h$ for some c, h, k chosen from a finite set. Hence $z = \eta x^{k/h}$ for a suitable choice of a root of a and a suitable fixed η . In the proof of Theorem 3.1 we have defined $p(X) = F(X^h, \eta X^k)$. Putting all together we find that

$$y = F(x, z) = F(x, \eta x^{k/h}) = p(x^{1/h}) = p(a^{m/h})$$

□

4 More general S -units

The possibility of an extension of the Main Theorem of [CZ00] to more general S -units was hinted in Remark 1 of that paper. We give here a precise statement and a proof.

Theorem 4.1. *Let $f(X, Y) = a_0(X)Y^d + a_1(X)Y^{d-1} + \cdots + a_d(X)$ be a polynomial of degree $d \geq 2$ in Y with coefficients in a number field k . Suppose that*

- (i) a_0 is constant
- (ii) the equation $f(0, Y)$ has no repeated roots

Let S be a finite set of places of k containing the archimedean ones. Also, fix an absolute value $|\cdot|$. Suppose that the equation

$$f(x, y) = z$$

has an infinite sequence of solutions $(x, y, z) \in O_S^ \times O_S \times O_S^*$, such that $\max(|x|, |z|) \rightarrow 0$ and such that*

- (iii) *the height of the S -units x and z are bounded in the following way for some constant $c > 0$*

$$h(x) < -c \log |x|$$

$$h(z) < -c \log |z|$$

then there exist an integer $h > 0$ and a polynomial $p(X) \in \bar{k}[X]$ such that $f(X^h, p(X))$ has only one term.

Remark 4.2. The conclusion is somehow optimal. If $f(X^h, p(X)) = \eta X^k$ then it is easy to find infinitely many solutions as in the previous theorem to the equation $f(x, y) = z$ for a suitable extension of k and a finite set of places S . In fact, consider a number field k such that $\eta \in k$ and $p(X) \in k[X]$. Fix an absolute value $|\cdot|$ and a number a such that $|a| < 1$. Take S so that η and a are S -units, and the coefficients of p are S -integers. If we put $x = a^{nh}$ and $z = \eta a^{nk}$ for varying $n \in \mathbb{N}$, then x and z are S -units satisfying (iii) of (4.1). We $y = p(a^n)$ an S -integer, then $f(x, y) = z$.

Theorem 4.1 easily implies part of Theorem 3.1 (except the statement on multiplicative dependence of course). I have preferred to prove first the version with powers to show how the proof contained in [CZ00] can be adapted to the case of number fields. Since the proof is very similar to that of Theorem 3.1, to avoid unnecessary repetitions, I will only point out the modifications needed to the previous argument, at the cost of appearing a bit sketchy on the conclusion. I recommend comparing the proof case by case with the proof of Theorem 3.1.

4.1 The proof

Proof. We write $|\cdot|_w$ for the place satisfying (iii). In analogy to the perfect power case, we put $m = -\log |x|_w$ and $n = -\log |z|_w$, not necessarily integers. The equation $f(x, y) = z$ has an infinite sequence of solutions (x, y, z) with $\min(m, n) \rightarrow \infty$. The sequence m/n has a subsequence convergent to a value in $[0, +\infty]$. We consider the solutions in such a subsequence and we subdivide cases according to the value of the limit.

We remark once and for all that the condition $h(x) < -c \log |x|_w$ implies

$$\sum_{v \in M_k} \log \max(1, |x|_v) = h(x) < cm$$

with each addend on the left a positive real number. Hence, for every $v \in M_k$

$$\log |x|_v \leq \log \max(1, |x|_v) < cm$$

So, if (iii) holds, then

$$|x|_v < C^m \tag{45}$$

$$|z|_v < C^n \tag{46}$$

for some $C > 1$ and for every place $v \in M_k$.

4.1.1 First case: $\lim m/n = 0$

Suppose that an absolute value $|\cdot|$ is such that $|z| \gg C^n$ for some $C > 1$ and an infinite sequence of solutions. Put $n' = \log |z| \gg n$. First, note that $|y|$ is unbounded in any sequence where n tends to $+\infty$. In fact, suppose that $|y|$ is bounded, then using that $m < \epsilon n$ for every $\epsilon > 0$ definitely and (45), we see that $|f(x, y)| \ll |x|^D \ll C^{Dm} \ll C^{\epsilon n}$, for some $D > 0$, while $|f(x, y)| = |z| \gg C^n$.

Let $\Delta \in \mathbb{Z}$ be a common denominator for the coefficients of $a_1(X)/d$. We can suppose $a_1(X) = 0$, in fact we can change variables replacing Y with $Y/\Delta - a_1(X)/(a_0 d)$ as in Theorem 3.1.

Since O_S^*/O_S^{*d} is finite (O_S^* is a finitely generated abelian group, Theorem 2.4), there exists an S -unit b such that for infinitely many solutions we have $z = bt^d$ for some S -unit t dependent on z . We divide both sides of the equation $f(a^m, y) = b^n$ by by^d to get

$$\left(\frac{t}{y}\right)^d - \frac{a_0}{b} = b^{-1}[a_2(x)y^{-2} + \cdots + a_d(x)y^{-d}] \tag{47}$$

Put $l = \log |t|$ and observe that $\lim m/l = \lim dm/n' = 0$. Recall that $|y| > 1$ definitely, moreover $|x| \ll C^m$ by (45) then there exists a constant $D > 0$ such that

$$|b^{-1}[a_2(x)y^{-2} + \cdots + a_d(x)y^{-d}]| \ll C^{mD} \quad (48)$$

Also note that for every $\epsilon > 0$ we have $C^{mD} \ll |t|^\epsilon$ because $|t|$ is asymptotic to $|z|^{1/d} \gg C^n$. Hence from (48)

$$|b^{-1}[a_2(x)y^{-2} + \cdots + a_d(x)y^{-d}]| \ll |t|^\epsilon \quad (49)$$

and also the same quantity multiplied by y^2

$$|b^{-1}[a_2(x) + \cdots + a_d(x)y^{2-d}]| \ll |t|^\epsilon \quad (50)$$

Combining (47) and (49) we get

$$\left| \frac{t}{y} \right|^d \ll |t|^\epsilon$$

implying $|y| \gg |t|^{l-\epsilon'}$ for every $\epsilon' > 0$. Then using (50) we find from (47) that for every $\delta > 0$

$$\left| \left(\frac{t}{y} \right)^d - \frac{a_0}{b} \right| \ll |y|^{-2} |t|^\epsilon \ll |y|^{-2+\delta}$$

Note that a_0/b is constant, then for a suitable determination of its root (and possibly passing to a subsequence as in the proof of Theorem 3.1) we get

$$\left| \left(\frac{a_0}{b} \right)^{1/d} - \frac{t}{y} \right| \ll |y|^{-2+\delta} \quad (51)$$

By repeatedly taking subsequences of solutions we can suppose that for every place $v \in S$ either $|z|_v \gg C^n$ for some $C > 1$, or $|z|_v \ll C^n$ for every $C > 1$. Let S' be the subset of S of the places v such that $|z|_v \gg C^n$ for some $C > 1$. Note that S' is not empty. In fact, since $|z|_w$ is very small (converges to 0) by the product formula there must be some $v \in S$ such that $|z|_v$ is big. More precisely $|z|_v \gg C^n$ for some constant $C > 1$ (recall that $n = -\log |z|_w$) and we can extract a subsequence so that the place v for which this happen is always the same. So S' is not empty.

Then, the bound found in (51) can be obtained for every place in S' . This justifies the existence of an infinite sequence of solutions (x, y, z) such that for every $v \in S'$ there exists an algebraic number $\alpha_v \in \bar{\mathbb{Q}}$ such that putting $z = bt^d$ for some t , it holds

$$\left| \alpha_v - \frac{t}{y} \right| \ll |y|_v^{-2+\delta}$$

We have already noted that $|y|_v$ goes to infinity, then the previous inequality implies that $|t|_v$ is asymptotic to $|\alpha_v|_v \cdot |y|_v$. With this in mind we can also write

$$\left| \alpha_v - \frac{t}{y} \right|_v \ll |t|_v^{-2+\delta}$$

We apply the generalized Roth's theorem and conclude as in the first case of Theorem 3.1. In fact, those bounds which contains y, b^l, α_v of the first case of Theorem 3.1 are still valid here if b^l is substituted with t .

4.1.2 Second case: $0 < \lim m/n < \infty$

This case needs even less modifications to the proof done for Theorem 3.1.

Consider the completion k_w of k with respect to the place w . Observe that x and z converge to 0. Passing to a subsequence we can assume that y converges to a number $y_0 \in k_w$ (since k_w is locally compact and $|y|_w$ is bounded) which must satisfy $f(0, y_0) = 0$ by a continuity argument. Thanks to hypothesis (ii), there is a power series expansion converging to y for (x, z) in a neighborhood of $(0, 0)$ given by

$$y = \sum_{i,j \geq 0} c_{i,j} x^i z^j$$

where $c_{i,j}$ are numbers in k_w algebraic over k . Fix an integer $H > 0$. We consider the truncation

$$S_H(x, z) = \sum_{0 \leq i+j \leq H} c_{i,j} x^i z^j$$

The bounds discussed in the second case of Theorem 3.1 which contain x, y, z, m, n are still valid here.

4.1.3 Third case: $\lim m/n = +\infty$

We factor $f(X, Y)$, viewed as a polynomial in Y , in the algebraic closure of $k(X)$

$$f(X, Y) = a_0(Y - \alpha_1(X)) \dots (Y - \alpha_d(X))$$

where the algebraic functions $\alpha_1, \dots, \alpha_d$ admit a Puiseux expansion at infinity of the form

$$\alpha_i(X) = \sum_{j=-h}^{\infty} a_{i,j} X^{-j/e}$$

convergent at the points $x \in k$ such that $|x|$ is large enough.

We reduce to the case $e = 1$. Since O_S^*/O_S^{*e} is finite (O_S^* is a finitely generated abelian group, Theorem 2.4) there exists $b \in O_S^*$ such that for infinitely many z we have $z = bt^e$ for some $t \in O_S^*$ dependent on z . Replacing X by bX^e we obtain a new equation $f_1(t, y) = z$ with infinitely many solutions satisfying the hypotheses of Theorem 4.1. But this time $f_1(X, Y)$ has roots $\alpha_i(bX^e)$ which are Laurent series in X^{-1} . We can therefore suppose $e = 1$.

As before, we can suppose that for every place $v \in S$ either $|z|_v \gg C^n$ for some $C > 1$, or $|z|_v \ll C^n$ for every $C > 1$. Let S' be the subset of S of the places v such that $|z|_v \gg C^n$ for some $C > 1$. S' is not empty.

As in the third case of the proof of Theorem 3.1 we can show that for every $v \in S'$ there exist numbers $0 < t_v < d$ and $1 \leq i(v) \leq d$ such that for every N large enough we have

$$\left| y - \sum_{j=-h}^{N-h-1} a_{i(v),j} x^{-j} \right|_v \ll |x|_v^{(\delta-h(d-t_v))/t_v} \quad (52)$$

holding for each solution we are considering and we conclude in the same way as in the third case of the proof of Theorem 3.1, in fact those bounds involving x, y, m, n are still valid here. □

Remark 4.3. Theorem 4.1 shows that in Theorem 3.1 we can take $a, b \in k^*$, not necessarily integers.

4.2 A corollary

The following corollary is a natural extension to number fields of the Corollary of [CZ00]. Note however that here, for simplicity, we only claim the existence of a rational function $p(X)$.

Corollary 4.4. *Let k be a number field. Let $d \geq 2$ be an integer, $u(X) \in k[X]$ be any polynomial, and $a \in k^*$ be a nonzero number, not root of unity. If the equation*

$$y^d = u(a^m) + a^n$$

has infinitely many integral solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$, then there exist an integer $h > 0$ and a rational function $p(X) \in \bar{k}[X, X^{-1}]$ such that $p(X)^d - u(X^h)$ has just one term.

Theorems 3.1 or 4.1 can be applied directly to the above equation only when $u(0) \neq 0$ because we need hypothesis (ii). The point is proving it when $u(0) = 0$. It

could be done using Theorem 4.1 as done (for $k = \mathbb{Q}$) in [CZ00]. However, the proof is quite long. We prefer to postpone it to the last chapter where a more general statement will be very easily proved as a consequence of Theorem 6.8. We state this corollary here since it will be used in the proof of Corollary 5.6 of the next section.

5 Modifying hypothesis (iii)

Hypothesis (iii) of Theorem 3.1 is that there exists an absolute value $|\cdot|$ of k such that $|a| < 1$ and $|b| < 1$. In the case $k = \mathbb{Q}$ and a, b integers it is equivalent to a and b not coprime. It was substantially used only in the second case ($0 < \lim m/n < \infty$). We needed it to have a completion of k where both a^m and b^n converge to 0 so that we could apply the implicit function theorem.

If we remove this hypothesis we need a different way to construct a series. We can say something only for a very special class of equations, for which we are able to write a power series using the Taylor expansion of the d -th root around 1. This is nothing else than a particular case of the implicit function theorem, however it is not directly applied to the original equation, first we need to rewrite the equation taking out the biggest order term.

Theorem 5.1. *Let u be a polynomial of degree $r > 0$ with coefficients in a number field k . Suppose that $u(0) = 0$ and let $k \geq 1$ be its order of zero. Let $d \geq 2$ be an integer. Fix $a, b \in k^*$ two nonzero numbers, not roots of unity. Suppose that the equation*

$$y^d = u(a^m) + b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$ and the limit $\lim m/n \in (0, +\infty)$ exists nonzero and finite. Suppose that either there exists an absolute value $|\cdot|$ of k such that $|a| < 1$ and

$$\lim \frac{m}{n} \neq \frac{\log |b|}{k \log |a|}$$

or there exists an absolute value $|\cdot|$ of k such that $|a| > 1$ and

$$\lim \frac{m}{n} \neq \frac{\log |b|}{r \log |a|}$$

Then there exist an integer $h > 0$ and a rational function $p(X) \in \bar{k}[X, X^{-1}]$ such that $p(X)^d - u(X^h)$ has only one term. Moreover, a and b are multiplicatively dependent.

Remark 5.2. We are supposing $u(0) = 0$, hence no constant term is admitted in the equation.

5.1 A proposition

We introduce a proposition which will reduce the proof of Theorem 5.1 to finding a suitable series for y . We have already met a similar statement in the second case of the proof of Theorem 3.1.

Proposition 5.3. *Let k be a number field and w be a place of k . Let S be a finite set of places of k containing all the archimedean ones. Fix two nonzero numbers a and b . Let $f \in k[X, Y]$ be monic¹ of degree $d \geq 2$ as a polynomial in Y . Suppose that the equation*

$$f(a^m, y) = b^n \quad (53)$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_S$ with $\min(m, n) \rightarrow \infty$ and $0 < \lim m/n < \infty$. Fix an integer $k \geq 1$ and suppose that there exist rational monomials $g_1, \dots, g_k \in k_w[X, X^{-1}, Z, Z^{-1}]$ with coefficients algebraic over $\mathbb{Q}$ and a Laurent power series $F \in k_w((X_1, \dots, X_k))$ with coefficients algebraic over $\mathbb{Q}$ absolutely convergent in a neighborhood of $0 \in k_w^k$ except at most in 0 , such that for infinitely many solutions of (53) we have

$$(i) \quad |g_1(a^m, b^n)|_w \ll C^{-n}, \dots, |g_k(a^m, b^n)|_w \ll C^{-n} \text{ for some } C > 1$$

$$(ii) \quad y = F(g_1(a^m, b^n), \dots, g_k(a^m, b^n)) \text{ if } n \text{ is large enough}$$

Then there exists an integer $h > 0$ and a rational function $p(X) \in \bar{k}[X, X^{-1}]$ such that $f(X^h, p(X))$ has only one term. Moreover a and b are multiplicatively dependent.

Proof. We fix H an integer and let n (that is the solution) vary. Later H will be chosen large enough. Let $F_H(X_1, \dots, X_k)$ be the series $F(X_1, \dots, X_k)$ truncated at the H -th term, that is it contains exactly the terms of total degree at most H of F . We have, for n large enough,

$$|F(g_1, \dots, g_k) - F_H(g_1, \dots, g_k)|_w \ll \theta^{nH}$$

because of hypotheses (i) and a variant of Lemma 3.6, for some $\theta < 1$. The lemma only applies to series in $k[[X]]$, here instead $F \in k((X))$ so the error term could be bigger of a factor C^{nD} where D is a constant. This is not a problem since we choose $H > D$ and possibly take a bigger $\theta < 1$. Then, using hypothesis (ii) we obtain

$$|y - F_H(g_1, \dots, g_k)|_w \ll \theta^{nH} \quad (54)$$

Let N be the number of monomials in k variables of total degree at most H . Call them $\beta_i(X_1, \dots, X_k)$ for $i = 1, \dots, N$. Then there exist numbers $b_i \in k_w$ algebraic over k such that

$$F_H(\mathbf{x}) = b_1\beta_1(\mathbf{x}) + \dots + b_N\beta_N(\mathbf{x}) \quad (55)$$

¹We will say that a polynomial in Y is monic in Y when the leading term is a power of Y times a constant in k , like in (i) of 3.1.

where $\mathbf{x}$ is the vector $(g_1, \dots, g_k)$. We possibly decrease N so that the expression (55) is minimal in the sense that if $c_i \in k_w$ are algebraic over k and $S_H(\mathbf{x}) = c_1\beta_1(\mathbf{x}) + \dots + c_N\beta_N(\mathbf{x})$ then no one of the c_i is zero. Substituting (55) in (54) we have

$$|y - b_1\beta_1(\mathbf{x}) + \dots + b_N\beta_N(\mathbf{x})|_w \ll C^{-nH} \quad (56)$$

Enlarging S we may suppose that $g_1(a^m, b^n), \dots, g_k(a^m, b^n)$ are all S -units (recall that the g_i are monomials). We now apply the Subspace Theorem 2.8 to points in O_S^{N+1} . For a solution (m, n, y) we put $P = (y, \beta_1(\mathbf{x}), \dots, \beta_N(\mathbf{x}))$. Note that $\mathbf{x}$ implicitly depends on the solution. We define linear forms

$$\begin{aligned} L_{i,v}(T) &= T_i && \text{if } i \neq 0 \text{ or } v \neq w \\ \text{and } L_{0,w}(T) &= T_0 - b_1T_1 - \dots - b_NT_N \end{aligned}$$

Doing so we have from (56)

$$|L_{0,w}(P)|_w \ll C^{-nH} \quad (57)$$

We would like to find $\epsilon > 0$ and an H big enough so that

$$\prod_{v \in S} \prod_{i=0}^N |L_{v,i}(P)|_v \ll H(P)^{-\epsilon}$$

for the sequence of solutions we are considering. The left hand side is

$$|L_{0,w}(P)|_w \left(\prod_{v \in S \setminus \{w\}} |y|_v \right) \left(\prod_{j=1}^N \prod_{v \in S} |c_j\beta_j|_v \right)$$

Since all the β_i are S -units, we just need

$$|L_{0,w}(P)|_w \left(\prod_{v \in S \setminus \{w\}} |y|_v \right) \ll H(P)^{-\epsilon} \quad (58)$$

We can achieve this for H large enough using the following three bounds

$$\begin{aligned} |L_{0,w}(P)|_w &\ll \theta^{nH} && \text{which is (57)} \\ |y|_v &\ll C^m && \forall v \in S \end{aligned} \quad (59)$$

$$H(P) \ll C^{mHDE} \quad (60)$$

for suitable constants $C > 1$ and $D, E > 0$. We now show these bounds.

We start observing that there exists a constant $C > 1$ such that $C^{-n} \ll |a^m|, |b^n| \ll C^n$ for any absolute value $|\cdot|$ because $0 < \lim m/n < \infty$. Hence $|\beta_i| \ll C^{nHE}$ for some constant $E > 0$ (dependent only on the g_i) because the β_i are monomials in a^m and b^n of total degree at most H . Also $|y| \ll C^n$ (which is (59)) since y satisfies $f(a^m, y) = b^n$ and f is monic in Y . We can now prove (60).

$$\begin{aligned} H(P) &= H(y, \beta_1, \dots, \beta_N) \\ &= \prod_{v \in M_k} \max(|y|_v, |\beta_1|_v, \dots, |\beta_N|_v) \\ &\leq \prod_{v \in S} \max(|y|_v, |\beta_1|_v, \dots, |\beta_N|_v) \quad \text{because } y, \beta_i \in O_S \\ &\ll C^{mHDE} \end{aligned}$$

Thanks to the bounds (57), (59) and (60) we see that we can choose H big enough and ϵ small enough so that (58) holds. Then the Subspace Theorem 2.8 tells us that infinitely many points are contained in a hyperplane, that is there exist $A_1, \dots, A_N \in k$ such that

$$A_0 y + A_1 \beta_1 + \dots + A_N \beta_N$$

A_0 is not zero otherwise we would find a non-trivial relation among the β_i contradicting the minimality of expression (55). Then

$$A_0 y = - \left(\frac{A_1}{A_0} \beta_1 + \dots + \frac{A_N}{A_0} \beta_N \right)$$

Since the β_i are rational monomials in a^m and b^n , we have proved the existence of a rational function $q(X, Y) \in k[X, X^{-1}, Y, Y^{-1}]$ such that

$$y = q(a^m, b^n)$$

Consider the polynomial $g(X, Y)$ given by the numerator of the rational function $f(X, q(X, Z)) - Z$. We want to prove $g \neq 0$. Suppose the contrary, that is $g = 0$, hence $f(X, p(X, Z)) = Z$. If $p(X, Z) = q(X, Z)/r(X, Z)$ then we multiply the equation $f(X, p(X, Z)) = Z$ by $r(X, Z)^d$ to obtain a polynomial equation

$$r(X, Z)^d f(X, p(X, Z)) = r(X, Z)^d Z$$

If $\deg_Z(q) \neq \deg_Z(r)$ the left hand side has degree in Z equal to $d \max(\deg_Z(q), \deg_Z(r))$ while the right hand side has degree in Z equal to $1 + d \cdot \deg_Z(r)$. Since $d \geq 2$ this is

not the case hence $\deg_Z(q) = \deg_Z(r)$. Then the left hand side has degree in Z less or equal to $d \cdot \deg_Z(r)$ while the right hand side has degree in Z equal to $1 + d \cdot \deg_Z(r)$ which is also not possible showing $g \neq 0$.

Since $g(a^m, b^n) = 0$ for every solution we are considering, we can apply Lemma 2.17 to find that there exist integers $h > 0, k$ and a $\eta \in \bar{\mathbb{Q}}$ such that $g(X^h, \eta X^k) = 0$. It means that $f(X^h, q(X^h, \eta X^k)) = \eta X^k$. Put $p(X) = q(X^h, \eta X^k)$ and we have the wanted statement. $\square$

5.2 The proof of the theorem

Proof. We have to prove the two different statements of Theorem 5.1. Each will be subdivided into two cases. The point is to find a suitable series, then we conclude with Proposition 5.3.

5.2.1 The statement where $|a| < 1$

Suppose that $|\cdot|$ is an absolute value such that $|a| < 1$ and

$$\lim \frac{m}{n} \neq \frac{\log |b|}{k \log |a|}$$

We have two cases.

First case: $\lim \frac{m}{n} < \frac{\log |b|}{k \log |a|}$. There exists $\epsilon > 0$ such that definitely

$$\frac{m}{n} < \frac{\log |b|}{k \log |a|} - \epsilon$$

Hence, multiplying by $nk \log |a|$ and taking $\exp$

$$|a|^{km} > |b|^n |a|^{-\epsilon nk} = |b|^n C^n \quad (61)$$

for some $C > 1$. If $u(x) = a_r x^r + \dots + a_k x^k$ we can rewrite the equation as

$$\begin{aligned} y^d &= a_k a^{mk} \left(1 + \frac{a_{k+1}}{a_k} a^m + \dots + \frac{a_r}{a_k} a^{m(r-k)} + \frac{b^n}{a_k a^{km}} \right) \\ y^d &= a_k a^{mk} (1 + x_1 + \dots + x_{r-k} + x_0) \end{aligned} \quad (62)$$

where

$$\begin{aligned} x_0 &= \frac{b^n}{a_k a^{km}} \quad \text{and} \\ x_i &= \frac{a_{k+i} a^{mi}}{a_k} \quad \text{if } i > 0 \end{aligned}$$

Then $|x_0| \ll C^{-n}$ by (61) and $|x_i| \ll |a|^{mi}$ for $i > 0$. Since $n \ll m \ll n$, by possibly decreasing $C > 1$, we obtain the following bound for every $i = 0, \dots, r - k$

$$|x_i| \ll C^{-n} \quad (63)$$

We use a Taylor expansion for $(1 + X)^{1/d}$ to get from (62)

$$y = (a_k a^{mk})^{1/d} \left(1 + \frac{1}{d}(x_0 + \dots + x_{r-k}) + \dots \right) \quad (64)$$

holding when the $|x_i|$ are all sufficiently small, where we choose a suitable d -th roots of unity. The coefficients of the series are numbers in the completion of k algebraic over k . Expression (64) is a series in the x_i convergent to y , absolutely convergent (because it is a Taylor series) in a neighbourhood of 0. The x_i are monomials in a^m and b^n satisfying (63). Hence, the conclusion is obtained from Proposition 5.3 applied with the number field $k(a^{1/d}, a_k^{1/d})$.

Second case: $\lim \frac{m}{n} > \frac{\log |b|}{k \log |a|}$. We work similarly to the previous case. There exists $\epsilon > 0$ such that definitely

$$\frac{m}{n} > \frac{\log |b|}{k \log |a|} + \epsilon$$

$$|a|^{km} < |b|^n |a|^{\epsilon nk} = |b|^n C^{-n} \quad (65)$$

for some $C > 1$. From $y^d = u(a^m) + b^n$ write

$$\begin{aligned} y^d &= b^n \left(1 + a_k \frac{a^{mk}}{b^n} + \dots + a_r \frac{a^{mr}}{b^n} \right) \\ y^d &= b^n (1 + x_0 + x_1 + \dots + x_{r-k}) \end{aligned}$$

where

$$x_i = \frac{a_{k+i} a^{m(k+i)}}{b^n}$$

Then $|x_i| \ll C^{-n} |a|^i$ by (65). Since $|a| < 1$ we have the following bound for every $i = 0, \dots, r - k$

$$|x_i| \ll C^{-n} \quad (66)$$

From now on we work as before using a Taylor expansion of the form

$$y = b^{n/d} \left(1 + \frac{1}{d}(x_0 + \dots + x_{r-k}) + \dots \right) \quad (67)$$

Expression (67) is a series in the x_i convergent to y . It is absolutely convergent on a neighbourhood of 0. The x_i are monomials in a^m and b^n satisfying (66). Hence, the conclusion is obtained from Proposition 5.3 applied with the number field $k(b^{1/d})$.

5.2.2 The statement where $|a| > 1$

Let's now move to the second claim. Let $|\cdot|$ be an absolute value such that $|a| > 1$ and suppose that

$$\lim \frac{m}{n} \neq \frac{\log |b|}{r \log |a|}$$

As before, we have two cases.

First case: $\lim \frac{m}{n} < \frac{\log |b|}{r \log |a|}$. Then there exists $\epsilon > 0$ such that definitely

$$\begin{aligned} \frac{m}{n} &< \frac{\log |b|}{r \log |a|} - \epsilon \\ |a|^{rm} &< |b|^n |a|^{-\epsilon nr} = |b|_w^n C^{-n} \end{aligned} \tag{68}$$

for some $C > 1$. Write

$$\begin{aligned} y^d &= b^n \left(1 + a_k \frac{a^{mk}}{b^n} + \cdots + a_r \frac{a^{mr}}{b^n} \right) \\ y^d &= b^n (1 + x_0 + \cdots + x_{r-k}) \end{aligned}$$

where

$$x_i = \frac{a_{k+i} a^{m(k+i)}}{b^n}$$

Then $|x_i| \ll C^{-n} |a|^{m(k+i-r)}$ by (68). Since $|a| > 1$ and $i \leq r-k$ we have $|a|^{m(k+i-r)} \leq 1$. Hence, we have the following bound for every $i = 0, \dots, r-k$

$$|x_i| \ll C^{-n} \tag{69}$$

From now on we work as before using a Taylor expansion of the form

$$y = b^{n/d} \left(1 + \frac{1}{d} (x_0 + \cdots + x_{r-k}) + \cdots \right) \tag{70}$$

Expression (70) is a series in the x_i . The x_i are monomials in a^m and b^n satisfying (69). Hence, the conclusion is obtained from Proposition 5.3 applied with the number field $k(b^{n/d})$.

Second case: $\lim \frac{m}{n} > \frac{\log |b|}{r \log |a|}$. There exists $\epsilon > 0$ such that definitely

$$\frac{m}{n} > \frac{\log |b|}{r \log |a|} + \epsilon$$

$$|a|^{rm} > |b|^n |a|^{\epsilon nk} = |b|_q^n C^n \quad (71)$$

for some $C > 0$. Write

$$\begin{aligned} y^d &= a_r a^{mr} \left(\frac{a_k}{a_r} a^{m(k-r)} + \dots + \frac{a_{r-1}}{a_r} a^{-m} + 1 + a_r \frac{b^n}{a_r a^{mr}} \right) \\ y^d &= a_r a^{mr} (1 + x_0 + \dots + x_{r-k}) \end{aligned}$$

where

$$\begin{aligned} x_0 &= \frac{b^n}{a_r a^{mr}} \quad \text{and} \\ x_i &= \frac{a_{r-i}}{a_r a^{mi}} \quad \text{if } i > 0 \end{aligned}$$

We note that $|x_0| \ll C^{-n}$ by (71) and $|x_i| \ll |a|^{-mi}$ for $i > 0$. Since $|a| > 1$ and $n \ll m \ll n$, we have the following bound for every i

$$|x_i| \ll C^{-n} \quad (72)$$

for a possibly smaller $C > 1$.

We proceed as before using a Taylor expansion of the form

$$y = (a_r a^{mr})^{1/d} \left(1 + \frac{1}{d} (x_0 + \dots + x_{r-k}) + \dots \right) \quad (73)$$

Expression (73) is a series in the x_i . The x_i are monomials in a^m and b^n satisfying (72). Hence, the conclusion is obtained from Proposition 5.3 applied with the number field $k(a^{1/d}, a_r^{1/d})$. $\square$

5.3 Corollaries

If a and b are multiplicatively independent we have the contronominal proposition. We state it since we believe it has a nicer form than Theorem 5.1.

Corollary 5.4. *Let u be a polynomial of degree $r > 0$ with coefficients in a number field k . Suppose that $u(0) = 0$ and let $k \geq 1$ be its order of zero. Let $d \geq 2$ be an integer. Fix $a, b \in k^*$ two nonzero numbers, not roots of unity. Assume that a and b are multiplicatively independent. If the equation*

$$y^d = u(a^m) + b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$ and if the limit $\lim m/n \in (0, +\infty)$ exists nonzero and finite, then for every absolute value $|\cdot|$ of k such that $|a| < 1$ we have

$$\lim \frac{m}{n} = \frac{\log |b|}{k \log |a|}$$

while if $|a| > 1$ we have

$$\lim \frac{m}{n} = \frac{\log |b|}{r \log |a|}$$

Remark 5.5. The elements of $O_k^* = O_{M_\infty}^*$ are by Definition 2.1 those whose norm is 1 for every non-archimedean absolute value. Suppose that $a \notin O_k^*$, then there exists a non-archimedean absolute value $|\cdot|$ such that $|a| \neq 1$. Note that $\log |b|/\log |a|$ is rational because $|\cdot|$ is non-archimedean. Hence, in the setting of Corollary 5.4, the limit of m/n must be *rational*.

In the case $k = \mathbb{Q}$ we have the following more complete statement. Note that we are never assuming a and b not coprime.

Corollary 5.6. *Let u be a non-constant polynomial with rational coefficients. Suppose that $u(0) = 0$. Let $d \geq 2$ be an integer. Fix $a, b \in \mathbb{Q}^*$ two nonzero rational numbers, different from 1 and -1 . Suppose that the equation*

$$y^d = u(a^m) + b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times \mathbb{Z}$ with $\min(m, n) \rightarrow \infty$. Then there exist an integer $h \neq 0$ and a rational function $p(X) \in \bar{\mathbb{Q}}[X, X^{-1}]$ such that $p(X)^d - u(X^h)$ has only one term. Moreover, a and b are multiplicatively dependent.

Proof. First, note that we can suppose that the limit of m/n exists. If it is 0 or ∞ we conclude as in the first or third case of Theorem 3.1 (see remarks 3.5 and 3.7). Hence, we can suppose $\lim m/n \in (0, \infty)$.

We now prove, by contradiction, that a and b must be multiplicatively dependent. Suppose not. Then we are in the hypothesis of Corollary 5.4 and as remarked in 5.5 the limit of m/n must be rational. Theorem 5.4 also tells us that the quantity $\log |b|/\log |a|$ is equal to $k \lim m/n$ or $r \lim m/n$ so it must be rational (here we are using the archimedean absolute value of $\mathbb{Q}$):

$$\frac{\log |b|}{\log |a|} \in \mathbb{Q}$$

This is possible only when a and b are multiplicatively dependent, but we were supposing the contrary. We have proved the claim.

Next, we need to find h and $p(X)$ such that $p(X)^d - u(X^h)$ has only one term. Since a and b are multiplicatively dependent $a^i = b^j$ for some integers $i, j > 0$, then there exists an integer $c > 1$ such that both a and b are powers of c . Hence, there are infinitely many solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times \mathbb{Z}$ with $\min(m, n) \rightarrow \infty$ to the equation

$$y^d = u(c^m) + c^n$$

and we obtain what we need using corollary 4.4. □

Example 5.7. For any integer $d \geq 2$ the equation

$$y^d = 2^n + 3^m$$

has only finitely many solutions $(m, n, y) \in \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$. In fact by Corollary 5.4 either m or n must be bounded (because 2 and 3 are multiplicatively independent) so that we may work as if m or n is fixed. Suppose, without loss of generality, that $n = n_0$ for infinitely many solutions. We have already discussed how to treat the case when there are only two unknowns: if there are infinitely many solutions, then $p(X)^d = X^h + 2^{n_0}$ for some $h > 0$ and $p(X) \in \bar{\mathbb{Q}}[X, X^{-1}]$ (Ex. 10 p.43 of [Zan03]). This is not possible.

An other equation (apparently more complicated) that can be treated in the same way is the following one.

Example 5.8. The equation

$$y^d = 2^n + 3^m + 4^n$$

has only finitely many solutions $(m, n, y) \in \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$.

6 Modifying hypothesis (ii)

Hypothesis (ii) of Theorem 3.1 is that $f(0, Y)$ has no repeated roots. It was needed to apply the implicit function theorem. In fact, if x, y, z satisfy $f(x, y) = z$ and y converges to some value y_0 , with (ii) we can write y as a series in x and z converging when $|x|$ and $|z|$ are small enough.

In this chapter, a way to weaken this hypothesis is presented. It requires looking at the Newton polygon of f . With some new hypotheses we are able to blow up the singularity, and after a change of variable, to write a series converging to y using, again, the implicit function theorem. Once we have the series the conclusion follows from Proposition 5.3.

This chapter is the core of my thesis.

6.1 The Newton polygon

We start by giving the important notion of the Newton polygon of a two-variables polynomials. It was originally introduced by Newton to prove the Newton-Puiseux theorem 2.13, see [Wal78] IV§3 for an explanation of this construction.

Definition 6.1. Let $f(X, Y) = \sum_{i,j \in \mathbb{R}} a_{i,j} X^i Y^j$ be a real-exponent polynomial, that is the exponents of X and Y are real and the sum is finite. We define the set

$$S_f = \{(i, j) | a_{i,j} \neq 0\} \subseteq \mathbb{R}^2$$

The *Newton polygon* of f is a polygon in the cartesian plane defined as the convex hull of the set S_f .

Definition 6.2. Let $f \in k[X, Y]$ be a polynomial

- We say that a side of the Newton polygon *delimits from below* if it is given by an equation $ax + by + c = 0$ with $a > 0$ and $ax + by + c \geq 0$ for any point $(x, y) \in S_f$. Note that the side cannot be horizontal.
- We say that a side of the Newton polygon *delimits from above* if it is given by an equation $ax + by + c = 0$ with $a > 0$ and $ax + by + c \leq 0$ for any point $(x, y) \in S_f$. Note that the side cannot be horizontal.

These concepts are useful since we have the following lemma.

Lemma 6.3. Let k be a normed field. Let $f(X, Y) \in k[X, Y]$ be a polynomial and $g : \mathbb{N} \rightarrow k^*$, $h : \mathbb{N} \rightarrow k^*$ be functions. Suppose that the quantity $\gamma := \lim \log |h| / \log |g| \in (-\infty, +\infty)$ exists finite and

$$f(g, h) = 0$$

- If $\lim |g| = 0$ then there is a side of the Newton polygon of f which delimits from below of slope $-\gamma^{-1}$.
- If $\lim |g| = \infty$ then there is a side of the Newton polygon of f which delimits from above of slope $-\gamma^{-1}$.

This statement makes it clear why we introduced Definition 6.2 but it is not exactly what we will need. We will use the following more general lemma, though a bit convoluted.

Lemma 6.4. *Let k be a normed field. Let $f(X, Y) \in k[X, Y]$ be a polynomial and $g : \mathbb{N} \rightarrow k^*$, $h : \mathbb{N} \rightarrow k^*$, $l : \mathbb{N} \rightarrow k^*$ be functions. Suppose that*

$$f(g, h) = l$$

and that both $\beta := \lim \log |l| / \log |g| \in (-\infty, +\infty)$ and $\gamma := \lim \log |h| / \log |g| \in (-\infty, +\infty)$ exist finite.

- *Suppose that X^β is not the smallest power of X contained in $f(X, Y)$ (i.e. cX^β is not the monomial with the smallest exponent in $f(X, 0)$). If $\lim |g| = 0$ then there is a side of the Newton polygon of $f(X, Y) - X^\beta$ which delimits from below of slope $-\gamma^{-1}$.*
- *Suppose that X^β is not the biggest power of X in $f(X, Y)$ (i.e. cX^β is not the monomial with the biggest exponent in $f(X, 0)$). If $\lim |g| = \infty$ then there is a side of the Newton polygon of $f(X, Y) - X^\beta$ which delimits from above of slope $-\gamma^{-1}$.*

Proof. Suppose that $\lim |g| = \infty$. Consider the set M of functions $k^3 \rightarrow k$ given by the union

$$M = \{m \in k^{(k^3)} \mid m(x, y, z) = m'(x, y) \text{ for some monomial } m' \text{ appearing in } f(X, Y)\} \\ \cup \{m(x, y, z) = -z\}$$

The equation $f(g, h) = l$ takes the form

$$\sum_{m \in M} m(g, h, l) = 0 \tag{74}$$

We assign to each function $m \in M$ a weight $w(m)$ as follows. If $m(X, Y, Z) = cX^aY^b$ is a monomial of f then $w(m) = a + b\gamma$. If $m(X, Y, Z) = -Z$ then $w(m) = \beta$. This cumbersome notation turns now useful since we have the uniform property

$$\log |m(f, g, l)| \text{ is asymptotic to } w(m) \log |g| \tag{75}$$

Suppose by contradiction that there is no side of the Newton polygon of $f(X, Y) - X^\beta$ which delimits from above of slope $-\gamma^{-1}$. Then a function $m_0 \in M$, has weight strictly bigger than any other function in M (here we use that X^β is not the biggest power of X in $f(X, Y)$). The equation (74) is

$$m_0(g, h, l) = - \sum_{m \in M \setminus \{m_0\}} m(g, h, l)$$

We take the $\log |\cdot|$ on both sides

$$\log |m_0(g, h, l)| = \log \left| \sum_{m \in M \setminus \{m_0\}} m(g, h, l) \right| \quad (76)$$

$$\leq \log \sum_{m \in M \setminus \{m_0\}} |m(g, h, l)| \quad (77)$$

The quantity (77) is at most

$$\log \left(\max_{m \in M \setminus \{m_0\}} |m(g, h, l)| \right) + \log(\#M - 1)$$

which, by property (75), is asymptotic to (use that $\log |g| \rightarrow \infty$)

$$\log |g| \max_{m \in M \setminus \{m_0\}} w(m)$$

The left hand side of (76), by property (75), is instead asymptotic to

$$\log |g| w(m_0)$$

Hence $\log |g| w(m_0)$ is asymptotic at most to $\log |g| \max_{m \in M \setminus \{m_0\}} w(m)$, which is not possible since m_0 has strictly maximal weight among the functions in M and $\log |g| > 0$ definitely. We have proved the claim.

When $\lim |g| = \infty$ the proof is the same but we should look at the smallest weight and hence consider only sides of the Newton polygon of $f(X, Y) - X^\beta$ which delimit from below. $\square$

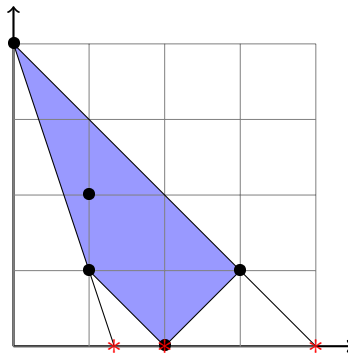
We introduce some other definitions. They are necessary to formulate the successive theorems.

Definition 6.5. We say that a polynomial $f \in k[X, Y]$ has the *Newton regularity property (NRP)* if no three points of S_f lie on a same side of the Newton polygon of f of negative slope delimiting from below or on a side delimiting from above.

Definition 6.6. Let $f \in k[X, Y]$ be a polynomial

- We define the set $C_f \subseteq \mathbb{Q}^+$ of the x -intercept of those sides of the Newton polygon of f which have negative slope and delimit from below.
- We also define the set $D_f \subseteq \mathbb{Q}$ of the x -intercept of those sides of the Newton polygon of f which delimit from above.
- Finally $C_f^* \subseteq \mathbb{Q}^+$ is the union of the sets C_{g_α} for all the multiple roots α of $f(0, Y)$, where $g_\alpha(X, Y) = f(X, Y - \alpha)$.

Example 6.7. If $f(X, Y) = Y^4 + XY^2 + X^3Y + XY + X^2$ then f has the NRP:



In this example $C_f = C_f^* = \{4/3, 2\}$ and $D_f = \{2, 4\}$.

6.2 The theorem

We want to prove, with a suitable hypothesis on the Newton polygon of f , that our equation $f(a^m, y) = b^n$ can have an infinite sequence of integral solutions with $\min(m, n) \rightarrow \infty$ only if the limit of m/n falls inside an explicit finite set of possible values.

Theorem 6.8. *Let $f \in k[X, Y]$ be a polynomial, monic in Y , of degree $d \geq 2$ in Y . Fix $a, b \in k^*$ two nonzero numbers, not roots of unity. Suppose that the equation*

$$f(a^m, y) = b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$. Suppose that the limit of n/m exists in $(0, \infty)$ and that either

- for any repeated root $y_0 \in \bar{k}$ of $f(0, Y)$ the polynomial $g(X, Y) = f(X, Y - y_0)$ has the NRP (see Definition 6.5), and for an absolute value $|\cdot|$ of k such that $|a| < 1$ and $|b| < 1$ we have (see Definition 6.6)

$$\frac{\log |b|}{\log |a|} \lim \frac{n}{m} \notin C_f^*$$

- or f has the NRP, and for an absolute value $|\cdot|$ of k such that $|a| > 1$ we have

$$\frac{\log |b|}{\log |a|} \lim \frac{n}{m} \notin D_f$$

Then there exist an integer $h > 0$ and a rational function $p(X) \in \bar{k}[X, X^{-1}]$ such that $f(X^h, p(X))$ has only one term. Moreover, a and b are multiplicatively dependent.

Remark 6.9. For a solution (m, n, y) we ask $y \in O_k$ for simplicity, however the following argument only requires $y \in O_S$ for some S .

6.3 The proof

Proof. Our goal is to write a series $F(X_1, \dots, X_k)$ such that $y = F(g_1(x, z), \dots, g_k(x, z))$ where $g_1, \dots, g_k$ are rational monomials in x and z such that $|g_i(x, z)| < C^{-n}$ for some constant $C > 1$. Then, we conclude using Proposition 5.3. We have already used this technique for example in the second case of Theorem 3.1. In that situation it was easy to apply the implicit function theorem. Here we do the same but we first need to make a change of variables to bypass the singularity.

We have to prove two different statements.

6.3.1 The statement $|a| < 1$ and $|b| < 1$

Suppose first that $|a| < 1$ and $|b| < 1$, and for any repeated root $y_0 \in \bar{k}$ of $f(0, Y)$ the polynomial $g(X, Y) = f(X, Y - y_0)$ has the NRP. We put $x = a^m$, $z = b^n$, then x and z converge to 0 in the completion k_v of k with respect to the given absolute value. We take a subsequence of solutions such that y converges to a root y_0 of the polynomial $f(0, Y)$ (in fact $|y|$ is bounded since $|a| < 1$ and $|b| < 1$ and y satisfies $f(x, y) = z$). The hypotheses and the conclusion are invariant by a change of variables $Y \rightarrow Y - y_0$ (we possibly extend the field k and consider solutions with y an S -integer as in remark 6.9). We can therefore suppose $y_0 = 0$. If 0 is a simple root of $f(0, Y)$ we conclude as in the second case of the proof of Theorem 3.1. Then,

we can suppose that 0 is not a simple root, hence $f(X, Y)$ has the NRP. Also, note that $C_f \subseteq C_f^*$.

If the limit of n/m exists in $(0, \infty)$, we define

$$\beta = \frac{\log |b|}{\log |a|} \lim \frac{n}{m} > 0$$

so that $\log |z|$ is asymptotic to $\beta \log |x|$. We know, by hypothesis, that $\beta \notin C_f^*$ hence also $\beta \notin C_f$.

If $y = 0$ frequently the conclusion is trivial, hence suppose $y \neq 0$. We consider a subsequence of solutions such that the following limit exists

$$\gamma = \lim \frac{\log |y|}{\log |x|}$$

Note that $|y|$ is bounded and we have $|z - f(0, y)| = |f(x, y) - f(0, y)| \ll |x|$ and $|z| \ll |x|^{\beta/2}$, hence $f(0, y) \ll |x|^{\min(1, \beta/2)}$. Moreover if $r > 0$ is the multiplicity in 0 of $f(0, Y)$ then $|y|^r \ll |f(0, y)| \ll |y|^r$ because y converges to 0, hence $|y|^r \ll |x|^{\min(1, \beta/2)}$ from which $\gamma > 0$.

There exists a constant $C > 1$ such that $C^{-n} < |x|_v, |z|_v < C^n$ for every place v . Hence $|y|_v < C^n$ for some other $C > 1$ because y satisfies $f(x, y) = z$. Then, by the product formula and the fact that y is an S -integer we also see that $C^{-n} \ll |y|_v \ll C^n$ for some other $C > 1$ proving that $-\infty < \gamma < \infty$. Then

$$0 < \gamma < \infty \tag{78}$$

Since $\beta \notin C_f$ and f is monic, in particular X^β is not the smallest power of X in $f(X, Y)$ (i.e. cX^β is not the monomial with the smallest exponent in $f(X, 0)$). Moreover $f(x, y) = z$ and $\log |z|$ is asymptotic to $\beta \log |x|$ while $\log |y|$ is asymptotic to $\gamma \log |x|$. Hence, by Lemma 6.4, a side of the Newton polygon of $f(X, Y) - X^\beta$ which delimits from below must have slope equal to $-\gamma^{-1}$. We observe that the side has negative slope by (78).

We subdivide two cases.

First case: the point $(\beta, 0)$ corresponding to the monomial X^β is not part of the side of slope $-\gamma^{-1}$ of the Newton polygon of $f(X, Y) - X^\beta$. For an intuition, this means that z is not a dominant term of the equation when $|x|, |z| \rightarrow 0$. We have that γ is rational since X^β is not part of the side of slope $-\gamma^{-1}$ of the Newton polygon (β is the only exponent of a monomial in $f(X, Y) - X^\beta$ which may be not rational).

We can write

$$y = x^\gamma y' \tag{79}$$

with y' a number such that $|x|^c \ll |y'| \ll |x|^{-c}$ for any $c > 0$ definitely. Here and in the following when taking a rational power we are always implicitly choosing a suitable root. Let's substitute (79) in the equation $f(x, y) = z$

$$f(x, x^\gamma y') = z$$

Let e be the x -intercept of the side we are considering, then the previous equation can be arranged in the following way

$$x^e b(y') + g(x, y') = z \quad (80)$$

where $b(Y)$ and $g(X, Y)$ are polynomials with the property that if $cX^\alpha Y$ is a monomial of g , then $\alpha > e$. Also, b is a *binomial*. In fact, on the side of slope $-\gamma^{-1}$ of the Newton polygon of $f(X, Y) - X^\beta$ there are only two monomials since X^β isn't there and f has the NRP. This means that the linear form $x_1 + \gamma x_2$ is minimized among the points $(x_1, x_2) \in S_f$ exactly at the two points on the side we are considering, where it has value e since the side is given by the equation $x_1 + \gamma x_2 = e$. Also note that $e < \beta$ in this first case. We divide both sides of (80) by x^e and we arrange the terms so that the equation becomes

$$b(y') + \sum_{i=1}^k g_i(x, z) t_i(y') = 0 \quad (81)$$

for some rational monomials g_i and polynomials t_i such that if $g_i(X, Z) = cX^{\alpha_1} Z^{\alpha_2}$ then $\alpha_1 + \beta\alpha_2 > 0$: actually, they all depend only on X except one which is $-X^{-e}Z$, recall that $-e + \beta > 0$. Hence $g_i(x, z) < C^{-n}$ because $\log |g_i(x, z)|$ is asymptotic to $(\alpha_1 + \beta\alpha_2) \log |x|$, in particular the $g_i(x, z)$ all converge to 0.

On a suitable subsequence, y' must converge to a nonzero value. In fact, suppose first that $|y'|$ converges to 0. Then, if $r > 0$ is the zero-multiplicity in 0 of $b(Y)$ we have $|y'|^r \ll |b(y')| \ll |y'|^r$. With the bounds $|x|^c \ll |y'| \ll |x|^{-c}$ holding for each $c > 0$, we find that $\log |b(y')|$ is asymptotic to 0, while it should be less than that because the other term of equation (81) has as logarithm of the module

$$\log \left| \sum_{i=1}^k g_i(x, z) t_i(y') \right| \ll \log \max_{1 \leq i \leq k} |g_i(x, z)|$$

which is less than 1 definitely. Similarly, if $|y'|$ goes to ∞ then $|y'|^d \ll |b(y')| \ll |y'|^d$ where d is the degree of $b(Y)$ and we find a contradiction as before. This shows that the limit of y' on a suitable subsequence exists nonzero.

Call y_0 the limit of y' . From equation (81) we can write y' as a series in $g_1, \dots, g_k$ converging when $|g_i|$ are all small enough thanks to the implicit function theorem. In fact, when substituting $g_1 = \dots = g_k = 0$ in (81) we find the polynomial $b(y') = 0$ and y_0 is a simple root of it since b is a binomial and y_0 is nonzero.

Since $y' = x^\gamma y'$ we also have a series converging to y of the wanted shape.

Second case: the point $(\beta, 0)$ corresponding to the monomial X^β is part of the side of slope $-\gamma^{-1}$ of the Newton polygon of $f(X, Y) - X^\beta$. Intuitively, this means that z is a dominant term of the equation when $|x|, |z| \rightarrow 0$. Since $\beta \notin C_f$ there are no three monomials aligned on the side of slope $-\gamma^{-1}$ of the Newton polygon of $f(X, Y) - X^\beta$.

One vertex of the Newton polygon contained in the side of slope $-\gamma^{-1}$ is $(\beta, 0)$, the other is (c_1, c_2) , say. Note that $c_2 > 0$ because the side is not horizontal. We can compute $\gamma = \frac{\beta - c_1}{c_2}$. Then

$$\log \left| \frac{z^{1/c_2}}{x^{c_1/c_2}} \right| = \frac{1}{c_2} \log \left| \frac{z}{x^{c_1}} \right| = \frac{1}{c_2} (\log |z| - c_1 \log |x|)$$

is asymptotic $\gamma \log |x|$ since $\log |z|$ is asymptotic to $\beta \log |x|$. We can rewrite y as

$$y = \frac{z^{1/c_2}}{x^{c_1/c_2}} y' \quad (82)$$

for a y' such that $|z|^c \ll |y'| \ll |z|^{-c}$ for any $c > 0$. Substituting (82) into the equation $f(x, y) = z$ we have

$$f \left(x, \frac{z^{1/c_2}}{x^{c_1/c_2}} y' \right) = z$$

which can be rewritten as

$$z(dy'^{c_2} - 1) = g(x, z, y') \quad (83)$$

where $g(X, Y, Z)$ is a polynomial such that every monomial $cX^{\alpha_1}Z^{\alpha_2}Y^{\alpha_3}$ appearing in it satisfies $\alpha_1 + \beta\alpha_2 > \beta$. Also, d is the coefficient of $X^{c_1}Y^{c_2}$ in $f(X, Y)$. This is because the side of the Newton polygon of $f(X, Y) - X^\beta$ delimits from below. In fact, this condition implies that the linear form $x_1 + \gamma x_2$ valued at the points $(x_1, x_2) \in S_f$ is minimized only at (c_1, c_2) where it has value $c_1 + \gamma c_2 = \beta$. By the linear change of variables $(x_1, x_2) \rightarrow (x_1 - \frac{c_1}{c_2}x_2, \frac{1}{c_2}x_2) = (\alpha_1, \alpha_2)$ the linear form becomes $\alpha_1 + \beta\alpha_2$. Note that $(c_1, c_2) \rightarrow (0, 1)$ so that the minimum is now reached in $(0, 1)$. This shows

that each monomial $cX^{\alpha_1}Z^{\alpha_2}$ in $f\left(X, \frac{Z^{1/c_2}}{X^{c_1/c_2}}\right)$ is such that $\alpha_1 + \beta\alpha_2 \geq \beta$ with equality holding only when $\alpha_1 = 0, \alpha_2 = 1$ corresponding to the monomial cZ .

We divide both sides of (83) by z , then there exist rational monomials $g_1, \dots, g_k$ and polynomials $t_1, \dots, t_k$ such that

$$(dy'^{c_2} - 1) = \sum_{i=1}^k g_i(x, z)t_i(y') \quad (84)$$

and if $g_i(X, Z) = cX^{\alpha_1}Z^{\alpha_2}$ then $\alpha_1 + \beta\alpha_2 > 0$. Hence $g_i(x, z) < C^{-n}$ because $\log |g_i(x, z)|$ is asymptotic to $(\alpha_1 + \beta\alpha_2) \log |x|$, in particular the $g_i(x, z)$ all converge to 0.

Exactly as in the previous case we see from (84) that $|y'|$ is bounded, then on a suitable subsequence y' converges to a value y_0 . With equation (84) we can write y as a series in $g_1, \dots, g_k$ converging when n is large enough thanks to the implicit function theorem. In fact when substituting $g_1 = \dots = g_k = 0$ in (84) we find the polynomial $dy'^{c_2} - 1$ (with $c_2 > 0$) which doesn't have multiple roots.

Since $y = \frac{z^{1/c_2}}{x^{c_1/c_2}}y'$ we also have a series converging to y of the wanted shape.

6.3.2 The statement where $|a| > 1$

Suppose that $|a| > 1$ and that f has the NRP. It works very similarly to the first statement.

We define

$$\beta = \frac{\log |b|}{\log |a|} \lim \frac{n}{m}$$

so that $\log |z|$ is asymptotic to $\beta \log |x|$. We know that $\beta \notin D_f$ and f has the NRP.

If $y = 0$ frequently the conclusion is trivial, hence suppose $y \neq 0$. We consider a subsequence of solutions such that the following limit exists

$$\gamma = \lim \frac{\log |y|}{\log |x|}$$

There exists a constant $C > 1$ such that $C^{-n} < |x|_v, |z|_v < C^n$ for every place v . Hence $|y|_v < C^n$ for some other $C > 1$ because y satisfies $f(x, y) = z$. Then, by the product formula and the fact that y is an S -integer we also see that $C^{-n} \ll |y|_v \ll C^n$ for some other $C > 1$ proving that

$$-\infty < \gamma < \infty$$

Since $\beta \notin D_f$, in particular X^β is not the biggest power of X in $f(X, 0)$ (i.e. cX^β is not the monomial with the biggest exponent in $f(X, 0)$). Moreover $f(x, y) = z$ and $\log |z|$ is asymptotic to $\beta \log |x|$ while $\log |y|$ is asymptotic to $\gamma \log |x|$. Hence, by Lemma 6.4, a side of the Newton polygon of $f(X, Y) - X^\beta$ which delimits from above must have slope equal to $-\gamma^{-1}$.

We subdivide as before two cases depending on whether or not the point $(\beta, 0)$ is part of the side of slope $-\gamma^{-1}$ of the Newton polygon of $f(X, Y) - X^\beta$. The proof given when $|a|, |b| < 1$ still works. Note that, in the previous statement, we used $|b| < 1$ only in one instance: when we proved $\gamma > 0$. Here we are not claiming that. We must reverse some inequality since now the side delimits from above. We define as before rational monomials $g_1, \dots, g_k$ such that $\log |g_i(x, z)|$ is asymptotic to $(\alpha_1 + \beta\alpha) \log |x|$. But now $\alpha_1 + \beta\alpha < 0$ while $\log |x| > 0$ so in any case $g_i(x, z) < C^{-n}$. $\square$

Remark 6.10. We required f to have the NRP to get to a binomial equation. Actually, we only need less. It is enough that the monomials on each given relevant side of S_f (those of negative slope which delimit from below and those which delimit from above) determine, after the substitution $X \rightarrow 1$, a polynomial with no multiple roots except at most 0.

6.4 Corollaries

The following proposition is the contronominal of Theorem 6.8. We state it since we believe it has a nicer form. It supposes a and b multiplicatively independent.

Corollary 6.11. *Let $f \in k[X, Y]$ be a polynomial, monic in Y , of degree $d \geq 2$ in Y which is not a perfect d -th power in $k[X, Y]$. Suppose that for any repeated root $y_0 \in \bar{k}$ of $f(0, Y)$ the polynomial $g(X, Y) = f(X, Y - y_0)$ has the NRP, and that f also has the NRP. Fix $a, b \in k^*$ two multiplicatively independent nonzero numbers, not roots of unity. Suppose that the equation*

$$f(a^m, y) = b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$. Suppose that the limit of m/n exists. Then for every absolute value $|\cdot|$ of k such that $|a| < 1$ and $|b| < 1$ we have

$$\frac{\log |b|}{\log |a|} \lim \frac{n}{m} \in C_f^*$$

Also, for every absolute value $|\cdot|$ of k such that $|a| > 1$ we have

$$\frac{\log |b|}{\log |a|} \lim \frac{n}{m} \in D_f$$

Proof. Since f is not a perfect d -th power in $\bar{k}[X, Y]$ we already know that $0 < \lim m/n < \infty$ (see remarks 3.5 and 3.7). Suppose by contradiction that the conclusion is not true. We can apply Theorem 6.8 to obtain a and b multiplicatively dependent against our hypothesis. $\square$

In the case $k = \mathbb{Q}$ we can state a corollary which has a shape very similar to that of Theorem 3.1.

Main Corollary 6.12. *Let $f(X, Y) = a_0(X)Y^d + a_1(X)Y^{d-1} + \dots + a_d(X)$ be a polynomial of degree $d \geq 2$ in Y with rational coefficients, not a perfect d -th power in $\bar{\mathbb{Q}}[X, Y]$. Let $a > 1$ and $b > 1$ be two integers. Suppose that*

(i)' a_0 is constant

(ii)' for any repeated root $y_0 \in \bar{k}$ of $f(0, Y)$, the polynomial $g(X, Y) = f(X, Y - y_0)$ has the NRP

(iii)' a and b are not coprime

If the equation

$$f(a^m, y) = b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$, then a, b are multiplicatively dependent.

Proof. If $f(0, Y)$ has no repeated roots we conclude with Theorem 3.1. If instead y_0 is a multiple root of $f(0, Y)$ by the change of variable $Y \rightarrow Y - y_0$ we can suppose $y_0 = 0$ so that (ii)' implies that f has the NRP.

We need to prove that a and b are multiplicatively dependent. Suppose the contrary. Then $\log |a| / \log |b|$ is not a rational number (this is the archimedean absolute value of $\mathbb{Q}$). Consider an infinite sequence of solutions such that $\lim m/n$ exists. Then $0 < \lim m/n < \infty$ by remarks 3.5 and 3.7 because we are supposing a and b multiplicatively independent.

Let p be a prime dividing both a and b , it exists by hypothesis (iii)'. If

$$\frac{\log |b|_p}{\log |a|_p} \lim \frac{n}{m} \notin C_f^*$$

then we get from Theorem 6.8 (first statement) that a and b are multiplicatively dependent against our assumption. Hence

$$\frac{\log |b|_p}{\log |a|_p} \lim \frac{n}{m} \in C_f^*$$

Then $\lim n/m$ must be a rational number. This implies that

$$\frac{\log |b|}{\log |a|} \lim \frac{n}{m} \notin D_f$$

because $\log |b|/\log |a|$ is not rational. Therefore we can again apply Theorem 6.8 (second statement) to get that a and b are multiplicatively dependent against our assumption. $\square$

Remark 6.13. Actually, in 6.12 we don't need $m, n > 0$ but only $\min(|m|, |n|) \rightarrow \infty$. In fact, an easy part of the proof of the Main Theorem of [CZ00] is to show that there is no infinite sequence of solutions with $\min(|m|, |n|) \rightarrow \infty$ when m or n is negative. Neither $(ii)'$ nor $(iii)'$ are needed.

Remark 6.14. The condition $(ii)'$ is strictly weaker than (ii) of Theorem 3.1. In fact it is always satisfied if $f(0, Y)$ doesn't have repeated roots. A polynomial which satisfies $(ii)'$ but not (ii) is $f(X, Y) = Y^2 + XY + X$. We then have the following example.

Example 6.15. The equation

$$y^2 + 2^n y + 2^n = 6^m$$

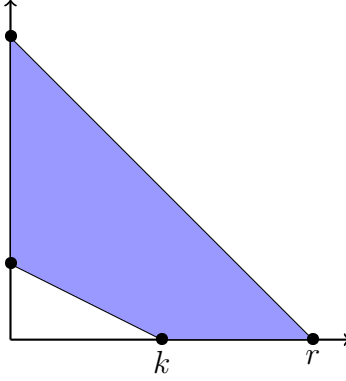
doesn't have an infinite sequence of solutions $(m, n, y) \in \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ with $\min(m, n) \rightarrow \infty$ because 2 and 6 are not coprime and multiplicatively independent. We use Corollary 6.12

Remark 6.16. We do not know if, under the hypotheses of 6.12, there exist an integer $h > 0$ and a polynomial $p(X) \in \bar{k}[X]$ such that $f(X^h, p(X))$ has only one term.

Remark 6.17. Hypothesis $(ii)'$ is always satisfied if $f(X, Y) = t(X) - u(Y)$ is a separated polynomial. Next section is devoted to this case.

6.5 The separated case

When $f(X, Y) = t(Y) - u(X)$ it is straightforward to calculate the sets C_f and D_f . Suppose that $u(x) = a_r x^r + \dots + a_k a^k$ has degree $r > 0$ and order of zero $k > 0$. For any root $y_0 \in \bar{k}$ of $f(0, Y)$ the Newton polygon of $f(X, Y - y_0)$ has the following shape.



Then f has the NRP and

$$C_f = C_f^* = \{k\} \quad \text{and} \quad D_f = \{r\}$$

Therefore, as an immediate consequence of Corollary 6.11 we have, for example, the following (and similarly we can rewrite Theorem 6.8 and Corollary 6.12)

Corollary 6.18. *Let t and u be polynomials with coefficients in a number field $\mathbf{k}$. Suppose that u has degree $r > 0$ and that t has degree $d \geq 2$. Suppose moreover that $u(0) = 0$ and let $k \geq 1$ be its order of zero. Fix $a, b \in \mathbf{k}^*$ two multiplicatively independent nonzero integers, not roots of unity. Suppose that the equation*

$$t(y) = u(a^m) + b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_{\mathbf{k}}$ with $\min(m, n) \rightarrow \infty$. Suppose that the limit of m/n exists. Then, for every absolute value $|\cdot|$ of $\mathbf{k}$ such that $|a| < 1$ and $|b| < 1$ we have

$$\lim \frac{m}{n} = \frac{\log |b|}{k \log |a|}$$

Also, for every absolute value $|\cdot|$ of $\mathbf{k}$ such that $|a| > 1$ we have

$$\lim \frac{m}{n} = \frac{\log |b|}{r \log |a|}$$

Remark 6.19. The hypothesis $u(0) = 0$ is harmless since we can always move a constant from u to t .

Remark 6.20. This corollary should be compared to Corollary 5.4. We have obtained the same exceptional sets by adopting a different point of view. The results are different: here we are considering more general equations but we are also supposing $|a| < 1$ and $|b| < 1$ *simultaneously*. If we were able to omit the condition $|b| < 1$ then we would find that there are only finitely many solutions to the equation $y^2 = 2^n + 3^m + c$, which is an open problem! See also the next example.

Example 6.21. Let $t(X) \in \mathbb{Q}[X]$ be a polynomial of degree at least 2, and c be an integer. If the equation

$$t(y) = 2^n + 3^m + c$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ with $\min(m, n) \rightarrow \infty$, then by Corollary 6.18

$$\lim \frac{m}{n} = \frac{\log 2}{\log 3}$$

In the special case $t(X) = X^d$, this was also proved in Ex. 13 p. 61 of [Zan03]. The method adopted there doesn't seem to extend to general t . It is not known, in general, whether or not this equation has infinitely many solutions.

As anticipated we can now state and prove a generalization of Corollary 4.4 which already appears (for the case $k = \mathbb{Q}$) in [CZ00]. The proof is much easier than the one contained in [CZ00].

Corollary 6.22. *Let t and u be polynomials with coefficients in a number field k . Suppose that t has degree $d \geq 2$ and that u is non constant. Fix $a \in k^*$ a nonzero number, not root of unity. If the equation*

$$t(y) = u(a^m) + a^n$$

has infinitely many integral solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$, then there exist an integer $h > 0$ and a rational function $p(X) \in \bar{k}[X, X^{-1}]$ such that $t(p(X)) - u(X^h)$ has just one term.

Proof. By remarks 3.5 and 3.7 on the proof of Theorem 3.1 if $\lim m/n = 0$ or $\lim m/n = \infty$ the conclusion holds. Note in fact that $t(Y) - u(X)$ is not a perfect d -th power in $\mathbb{C}[X, Y]$. Hence, we can assume $0 < \lim m/n < \infty$.

Moving a constant we suppose $u(0) = 0$. Let $r > 0$ and $k > 0$ be respectively the degree and the order of zero of $u(X)$. Suppose, by contradiction, that there are no h and $p(X)$ as wanted. Then, by Theorem 6.8 applied with the separated polynomial

$f(X, Y) = t(Y) - u(X)$, for every absolute value $|\cdot|$ of k such that $|a| < 1$ we must have

$$\lim \frac{m}{n} = \frac{\log |a|}{k \log |a|} = \frac{1}{k} \quad (85)$$

while if $|a| > 1$ we must have

$$\lim \frac{m}{n} = \frac{\log |a|}{r \log |a|} = \frac{1}{r} \quad (86)$$

Since a is not a root of unity there exist two absolute values $|\cdot|_1$ and $|\cdot|_2$ such that $|a|_1 > 1$ and $|a|_2 < 1$ by Theorem 2.2 and the product formula. Hence from (85) and (86) we get that $r = k$, that is u is a monomial. It is then trivial to find h and $p(X)$ such that $t(p(X)) - u(X^h)$ has just one term: take $p(X)$ constantly equal to a root of t . We have found a contradiction. $\square$

Let's now consider the case $k = \mathbb{Q}$. If a and b are not coprime we state one last corollary. It should be compared to Corollary 5.6. The difference is that here the equation is more general but a and b must be not coprime. It constitutes a different generalization of the Corollary of [CZ00] (now we assume $k = \mathbb{Q}$ but a is not necessarily equal to b).

Corollary 6.23. *Let t and u be polynomials with rational coefficients. Suppose that t has degree $d \geq 2$ and that u is not constant. Fix $a > 1$ and $b > 1$ two integers not coprime. If the equation*

$$t(y) = u(a^m) + b^n$$

has an infinite sequence of solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times O_k$ with $\min(m, n) \rightarrow \infty$, then there exist an integer $h > 0$ and a polynomial $p(X) \in \bar{\mathbb{Q}}[X]$ such that $t(p(X)) - u(X^h)$ has only one term. Moreover a and b are multiplicatively dependent.

Proof. Since a and b are not coprime and a separated polynomial has the NRP, we can apply Corollary 6.12 to find that a and b are multiplicatively dependent which is one claim. This means $a^i = b^j$ for some integers $i, j > 0$, then there exists an integer $c > 1$ such that both a and b are powers of c . Hence, there are infinitely many solutions $(m, n, y) \in \mathbb{N} \times \mathbb{N} \times \mathbb{Z}$ with $\min(m, n) \rightarrow \infty$ to the equation

$$t(y) = u(c^m) + c^n \quad (87)$$

and we obtain what we need using Corollary 6.22. $\square$

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